AI TRAFFIC-SAFETY CONSULTANT

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CONCEPT OF OPERATIONS
AI CORPS – TEAM 4

REVISION – C
1 October 2026

CONCEPT OF OPERATIONS
FOR
AI Traffic-Safety Consultant

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Change Record

Table 1 List of changes

Rev.	Date	Originator	Approvals	Description
-	14 Sep 2026	Juan Lopez, Jose Ruiz Garcia, Param Shelar, Kabir Singh		Draft release (AI Consulting Assistant).
A	16 Sep 2026	Juan Lopez, Jose Ruiz Garcia, Param Shelar, Kabir Singh		Added system architecture, subsystems, success criteria, operating modes, scenarios, legal and data constraints, and references.
B	20 Sep 2026	Juan Lopez, Jose Ruiz Garcia, Param Shelar, Kabir Singh		Added consulting-spend baseline, narrowed scope to one output class, evaluation method, go/no-go gate, team roles, risk register, and procurement and sustainment constraints.
C	1 Oct 2026	Juan Lopez, Jose Ruiz Garcia, Param Shelar, Kabir Singh		Re-scoped to one consultant deliverable with public ground truth: traffic-safety site prioritization and countermeasure memos (renamed AI Traffic-Safety Consultant). Re-baselined to four subsystems with named owners, aligned with the FSR, ICD, and Execution and Validation Plan. Reduced scope: chat page as a stretch goal, PDF-only memo export, one reference test host. Reformatted to the course template.

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1. Executive Summary

The City of College Station relies on roughly 40 paid outside consultants for work its staff lack the time or specialized expertise to complete. The AI Corps mandate for Team 4 is to test whether large language models (LLMs) can reduce that reliance. Revision B of this document proposed a general AI Consulting Assistant that drafted consultant narrative. Review of that concept showed that its ground truth was weak: awarded proposals may be withheld, the evaluation set was three to five documents, and quality could only be judged subjectively. Revision C narrows the project to one consultant deliverable that has public inputs, a public consultant product to compare against, and an objective score.

That deliverable is traffic-safety prioritization. The Bryan/College Station Metropolitan Planning Organization (MPO) Comprehensive Safety Action Plan, prepared by Kimley-Horn, includes a High-Injury Network (HIN) built from 2017–2022 crash history; a comparable consultant safety action plan costs about $200,000. The HIN ranks roads by past crash counts, a method that suffers from regression to the mean. The AI Traffic-Safety Consultant (ATSC) replaces that ranking with a machine-learning model corrected by the Empirical Bayes (EB) method, tests the result on crashes that happened after the consultant's data window closed, and drafts the countermeasure memo a consultant would write for each flagged site.

The ATSC has four subsystems, each with one owner:

- **S1 Road Network, Features, and Database** (Kabir Singh): turns public crash and roadway data into a validated, dated, site-level database.
- **S2 Risk Model and Backtest** (Param Shelar, also project leader): ranks every site by expected fatal and serious-injury (F&SI) crashes and proves or disproves the ranking against the HIN.
- **S3 Countermeasure Agent** (Jose Ruiz Garcia): drafts a cited countermeasure memo for each flagged site; deterministic tools compute every number and the LLM only writes and cites.
- **S4 Web Application and Deployment** (Juan Lopez): delivers the risk map, site pages, backtest results, and exports to City staff on a single low-cost host.

Success is an honest, measured answer rather than a promised result. The backtest reports capture rates with confidence intervals whatever they show, memo quality is measured by blind expert grading, and the cost of each memo is logged and set beside the consultant fee. Results go to City Hall in the fall and to City Council at the end of the spring 2027 semester. The Functional System Requirements (FSR), Interface Control Document (ICD), and Execution and Validation Plan that follow this document are derived from it.

2. Introduction

This Concept of Operations (ConOps) describes the problem the ATSC addresses, the proposed system and its subsystems, how City staff will use it, and its constraints and limitations. Details such as the final hosting location and the date of each City presentation will be confirmed with City staff.

2.1. Background

The City contracts outside consultants for engineering studies, planning, information technology, and other specialized services, paid from public funds within a FY2026 adopted budget of $474,225,698. Team 4's kickoff assignment was to find consultant work that LLMs can

reproduce at comparable quality and lower cost, and to measure the result honestly. Traffic safety is a strong test case for three reasons:

- **Clear need.** Brazos County recorded more than 20,000 crashes from 2017 to 2022, including 101 fatal and 524 serious-injury crashes. The MPO, Brazos County, Bryan, and College Station have adopted Vision Zero goals for 2035.
- **Public inputs.** Crash records (TxDOT CRIS), the TxDOT roadway inventory, OpenStreetMap, and MPO and City GIS layers are public, so no data acquisition depends on records requests.
- **Objective score.** A ranking made with data through 2022 can be scored on the crashes that actually occurred in 2023–2025. The consultant's HIN can be scored the same way, so the comparison is direct.

Ranking roads by raw crash counts overstates sites that had a few unlucky years; those sites tend to "improve" on their own. The Highway Safety Manual's Empirical Bayes method corrects this by blending each site's observed history with a model prediction for similar sites. The ATSC applies the same correction to a gradient-boosted model and tests whether the result finds future F&SI crashes better than the HIN does.

Federal Safe Streets and Roads for All (SS4A) grants were authorized under the Infrastructure Investment and Jobs Act through 30 September 2026, so future federal safety funding is uncertain. The ATSC is therefore framed as a prioritization tool for the City's own capital budget, not as a grant-writing aid.

2.2. Overview

City staff use one web application to see where severe crashes are most likely, why each site was flagged, what could be done about it, and how the ranking compares with the consultant's. Behind the application, the system ingests public data, builds a network of road segments and intersections, trains a risk model on a standard CPU, runs the backtest, and generates countermeasure memos through the TAMU AI Chat API. The team trains no LLMs; it uses existing models through the API. Figure 2 in Section 3.3 shows the system and its subsystems.

2.3. Referenced Documents and Standards

The following documents and standards apply to this concept. The FSR, ICD, and Execution and Validation Plan are the controlling documents for requirements, interfaces, and verification.

Table 2. Referenced Documents and Standards

Document	Revision / Date	Use in this project
ATSC Functional System Requirements	Rev. – (Draft), 30 Sep 2026	Binding requirements derived from this ConOps
ATSC Interface Control Document	Rev. – (Draft), 30 Sep 2026	Interfaces IF-01 to IF-12 and subsystem boundaries
ATSC Execution and Validation Plan	Rev. – (Draft), 30 Sep 2026	Schedule, ownership, verification of each requirement
B/CS MPO Comprehensive Safety Action Plan (Kimley-Horn)	Study began Oct 2023	Source of the High-Injury Network (comparison baseline)
MPO safety briefing	Jan 2026	Brazos County crash statistics; Safe System approach
AASHTO Highway Safety Manual	1st Ed., 2010 (2014 Suppl.)	Network screening, Empirical Bayes method, KABCO scale
FHWA Proven Safety Countermeasures	2021 update	Countermeasure library
FHWA CMF Clearinghouse	Online, accessed 2026	Crash modification factors and quality ratings

Document	Revision / Date	Use in this project
TxDOT CRIS	2015–2025 extracts	Crash records and data dictionary
TTI Crash Location Coordinates Tip Card	2022	Crash coordinate accuracy; map-matching tolerance
TAMU AI Chat API documentation (docs.tamus.ai)	Accessed 2026	LLM access, keys, and usage limits
NIST AI Risk Management Framework	AI RMF 1.0	Human review, logging, and disclosure practice
Texas Government Code, Ch. 552	Current	Public Information Act; outputs may be public records
W3C WCAG 2.1	June 2018	Web accessibility, Level AA
Zheng et al., Judging LLM-as-a-Judge (arXiv:2306.05685)	2023	Practice and biases of LLM-assisted grading

3. Operating Concept

3.1. Scope

The ATSC reproduces the analysis portion of a traffic-safety action plan for the roads inside College Station city limits. It performs four functions:

- **Risk ranking:** estimate expected F&SI crashes per mile per year for every road segment (about 0.1 mile) and intersection (any node where three or more road legs meet, with a 250 ft influence zone), and flag the highest-ranked sites covering 11% of network length, the same share the HIN covers.
- **Backtest:** rank sites using only information available by 31 December 2022, then measure the share of 2023–2025 F&SI crashes that occurred on flagged sites for the consultant HIN (method A), a negative binomial model with EB (method B, the industry standard), and the ATSC model (method C).
- **Countermeasure memos:** for each flagged site, a memo with the crash profile, the main risk factors, applicable FHWA Proven Safety Countermeasures with crash modification factors (CMFs) and sources, and a planning-level benefit/cost estimate.
- **Web application:** a risk map, site detail pages, a backtest page, and exports to CSV, GeoJSON, and PDF.

Stretch goal. A question-answering chat page is built only if the core functions are complete and verified by milestone M6; no binding requirement depends on it.

Out of scope. Public engagement, field studies, engineering design, and PE-sealed work remain with qualified engineers. The system makes no statements about individual drivers and is not a real-time operations tool. The consultant-spend map and RFP drafting functions of Revision B are removed from scope.

3.2. Operational Description and Constraints

The ATSC runs on an annual cycle, shown in Figure 1. When a new year of CRIS data is published, an operator refreshes the sources and runs the pipeline. The model ranks the sites, the backtest confirms that the method still holds, and memos are generated in a batch for the flagged sites. City staff then review the results in the web application, edit or approve memos, and export them for capital planning. All outputs are advisory until a qualified staff member approves them.

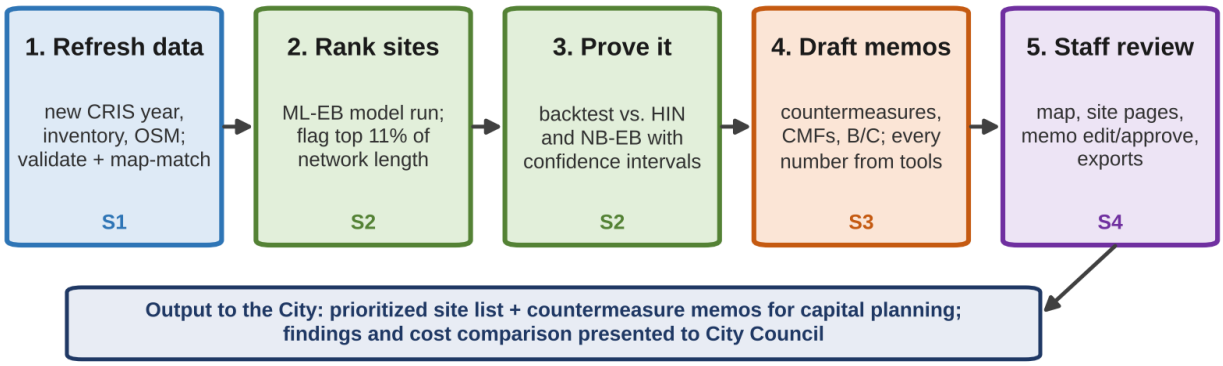


Figure 1. Operational Workflow and Owning Subsystem

The main operating constraints are:

- **Model access:** existing LLMs through the OpenAI-compatible TAMU AI Chat API with allocated tokens; keys are issued through chat.tamu.ai. No LLM is trained. The risk model is a gradient-boosted model that trains on a CPU.
- **Budget:** API charges are covered by the program; the team must obtain approval before exceeding the allocation. Every call is logged with its cost, with a target of $1.00 or less per memo.
- **Hosting:** the complete system runs on one Linux host with no more than 4 CPU cores, 8 GB of RAM, and 128 GB of storage, and no GPU. A Raspberry Pi 5 (ARM64) is the reference test host.
- **Data:** public sources only. The MPO HIN GIS layer must be requested from the MPO; if it is not received by 9 October 2026, the HIN is reconstructed from its published maps and the limitation is reported.
- **Data quality:** crash coordinates are noisy (in one TTI sample only 25% of officer and TxDOT coordinates agreed within 50 ft), so map-matching uses a generous, logged tolerance and must place at least 95% of geocoded F&SI crashes.
- **Temporal integrity:** every backtest input must be knowable as of 31 December 2022; an automated check stops any run that uses later information.
- **Data classification:** TAMU AI Chat is approved for data up to University-Confidential and does not train on inputs; the ATSC sends it only public site attributes, crash aggregates, and countermeasure records.
- **Public records:** outputs used in City business may become public records under the Public Information Act, so inputs, outputs, and model versions are logged.
- **Professional practice:** memos are planning-level drafts for review by qualified traffic engineers and do not replace an engineering study.
- **Sustainment:** the TAMU API allocation does not transfer to the City. After handoff the City supplies its own OpenAI-compatible endpoint and a host meeting the hosting constraint.
- **Schedule:** backtest results and the risk map are presented at City Hall in the fall; the full system and its quality and cost comparison are presented to City Council at the end of the spring 2027 semester.

3.3. System Description

The ATSC consists of four subsystems, shown in Figure 2 and described in Table 3. Each subsystem has one owner who is responsible for its design, its unit tests, and the FSR

requirements allocated to it. Subsystems exchange data only through the database interface (IF-09) and the REST API (IF-08) defined in the ICD, so each can be built and tested independently. Acquiring the source files is a one-time project task tracked in the Execution Plan (milestone M1), not a subsystem.

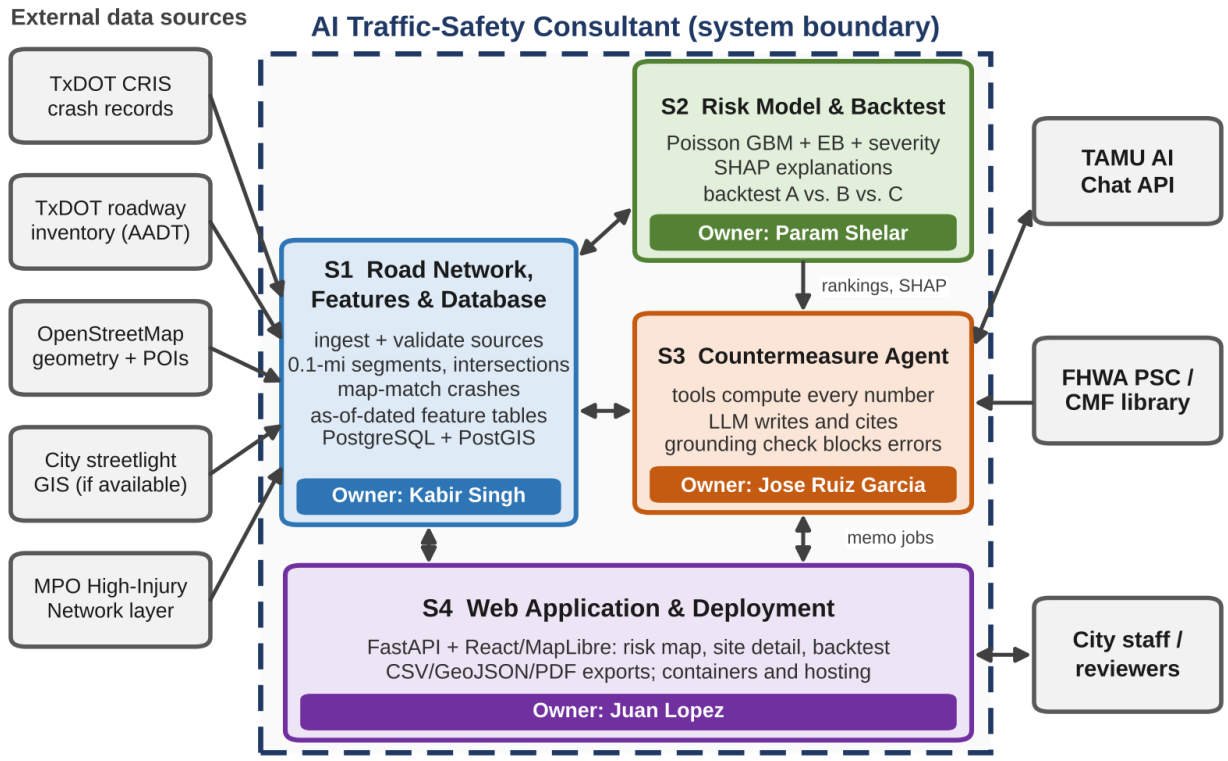


Figure 2. System Block Diagram with Subsystem Owners

Table 3. Subsystem Descriptions and Ownership

Subsystem	Owner	Function	Key engineering work
S1 Road Network, Features, and Database	Kabir Singh	Turn public data into a validated, dated, site-level dataset	Validate and load CRIS 2015–2025 and roadway/GIS sources; split the network into 0.1-mile segments and intersections (three or more legs, 250 ft influence zone); map-match crashes with a logged tolerance; compute features with as-of dates and enforce the leakage cutoff; design and operate the PostgreSQL/PostGIS database and model-run registry
S2 Risk Model and Backtest	Param Shelar	Rank sites by expected F&SI risk and prove the ranking	Poisson gradient-boosted model with overdispersion; EB weight $w = 1/(1 + k\Sigma\mu)$; severity model for P(K or A); SHAP explanations; negative binomial SPF baseline; flag 11% of network length; backtest of A, B, C with 1,000-resample bootstrap intervals; sensitivity at 5% and 20%; model card
S3 Countermeasure Agent	Jose Ruiz Garcia	Draft a consultant-style countermeasure memo for each flagged site	Curate the countermeasure library from FHWA PSC and the CMF Clearinghouse; four deterministic tools (crash profile, site context, countermeasure search, benefit/cost); prompts and memo template; numeric and citation grounding checks; cost and latency logging

Subsystem	Owner	Function	Key engineering work
S4 Web Application and Deployment	Juan Lopez	Put results in front of City staff and keep the system deployable	FastAPI REST API per the ICD; React/MapLibre risk map, site detail, backtest, and memo pages; CSV, GeoJSON, and PDF exports; WCAG 2.1 AA accessibility; containers, TLS, and deployment on the reference host

Success will be judged against the criteria in Table 4. Each criterion maps to a requirement in the FSR and a test in the Execution and Validation Plan.

Table 4. Success Criteria

Measure	Method	Target
Backtest capture rate	Share of 2023–2025 F&SI crashes on flagged sites (11% of network) for methods A, B, C; 1,000-resample bootstrap	Reported with 95% CI whatever it shows; goal: C above A
Map-matching	Geocoded F&SI crashes assigned to a site; unmatched crashes logged	At least 95%
Memo grounding	Seeded memos with invented numbers or missing citations	All blocked before publication
Memo quality	Blind grading of 15 sites by a traffic-safety reviewer on a six-criterion rubric	Mean ≥ 3.5 of 5 and within 0.5 of reference memos (goal)
Cost and time per memo	Logged tokens, cost, and latency for every memo	Median ≤ $1.00 and ≤ 2 minutes
Responsiveness	Scripted browser test, 5 concurrent users	Map ≤ 5 s; site page ≤ 2 s
User acceptance	Two to four City staff complete five tasks	≥ 80% of tasks unaided; written client acceptance

3.4. Modes of Operations

The system operates in the modes listed in Table 5. All user functions are available in normal mode.

Table 5. Modes of Operation

Mode	Description
Normal	All functions available; memos are generated through the API with every number traced to a tool result.
Degraded	The LLM API is unavailable or the token budget is reached. The risk map, site pages, and backtest stay available; new memo requests are marked "pending" until capacity returns.
Data refresh / model run	An operator loads new data and runs the pipeline from the command line. The last published run stays visible until the new run passes the built-in tests and is published.
Backtest	Inputs are frozen as of 31 December 2022 and the leakage check is enforced; results are written to the backtest page and the model card.
Administration	Maintainers update prompts, the countermeasure library, user roles, and budget thresholds, and can roll back to any stored model run. Changes are logged.

3.5. Users

The intended users are City of College Station staff. The team's primary City contact is Sam Rivera (Information Technology). User roles are summarized in Table 6.

Table 6. User Roles

Role	Responsibilities	Expected skill level
Planner / engineer (primary user)	Reviews ranked sites, reads memos, exports results for capital planning	Knows the City's roads; no technical training required
Reviewer	Checks memos and citations, edits or approves them, decides when an engineering study is needed	Traffic-safety expertise; short training on reviewing AI output
Administrator / operator	Runs the annual data refresh and model run, publishes or rolls back runs, manages keys and budget	City IT staff with basic scripting and Docker skills
City Council and MPO	Receive summary findings; do not use the system day to day	None

3.6. Support

During development, Team 4 meets with the City contact and the course T/A at least every other week. Support for the delivered system includes:

- **Documentation:** a user guide for City staff, a data dictionary, a model card, and an operator runbook for the annual data refresh.
- **Handoff:** source code, container images, configuration, the countermeasure library, and the evaluation package, plus a guide for pointing the system at a City-supplied LLM endpoint.
- **Monitoring:** logs of every pipeline stage, model run, LLM call, and API request, with built-in tests and an alert when API spending reaches 75% of the budget.
- **Training:** a short session for City staff before user acceptance testing.

Team responsibilities are assigned in Table 7. The project leader also owns one subsystem.

Table 7. Team Roles and Subsystem Ownership

Role	Member	Responsibility
Project leader	Param Shelar	Schedule, scope, City and course coordination, go/no-go decision (M2), integration, requirements verification matrix, risk register
S1 owner	Kabir Singh	Road network, map-matching, feature store, database; custodian of the ICD
S2 owner	Param Shelar	Risk model, EB correction, severity model, SHAP, backtest, model card
S3 owner	Jose Ruiz Garcia	Countermeasure library, agent tools, prompts, grounding checks, cost logging
S4 owner and City liaison	Juan Lopez	Web application, exports, deployment; City meetings, reviewer recruitment, blind grading and user acceptance sessions

3.7. Development Plan and Milestones

The fall semester delivers the data pipeline, risk model, and backtest; the spring semester delivers the agent, web application, and validation. Table 8 lists the milestones; the Execution and Validation Plan gives the full schedule, exit criteria, and go/no-go outcomes.

Table 8. Milestones

ID	Milestone	Target	Lead
M0	FSR, ICD, and Execution and Validation Plan submitted	1 Oct 2026	Param Shelar (all authors)
M1	Data acquired and validated	9 Oct 2026	Kabir Singh (S1)

ID	Milestone	Target	Lead
M2	Go/no-go: methods A, B, C run end to end	23 Oct 2026	Param Shelar (S2, lead)
M3	City Hall presentation	Nov 2026 (TBD)	Juan Lopez (liaison), all
M4	Model v1 frozen; fall report	11 Dec 2026	Param Shelar (S2)
M5	Agent MVP with grounding checks	19 Feb 2027	Jose Ruiz Garcia (S3)
M6	Web beta with exports	12 Mar 2027	Juan Lopez (S4)
M7	Validation complete	9 Apr 2027	Param Shelar (matrix), all owners
M8	City Council presentation	Apr 2027 (TBD)	All
M9	Final delivery and handover	7 May 2027	Param Shelar, all

4. Scenarios

4.1. Annual Network Screening

A City transportation planner preparing the next capital improvement program opens the risk map. Sites are colored by expected F&SI crashes per mile per year, and the flagged network is drawn with a heavier line. The planner toggles the HIN overlay to see where the model and the consultant agree and disagree, filters to the flagged sites, and exports the ranked list as CSV and GeoJSON for the City GIS. The list becomes the starting point for project selection, alongside cost and the City's own priorities.

4.2. Explaining a Flagged Site and Drafting a Memo

A City engineer asks why a four-lane arterial segment is ranked in the top 2%. The site page shows its crash history by year, type, and severity, the EB estimate, and the five factors that contributed most to the prediction, such as a 45 mph posted speed and no street lighting. The engineer opens the countermeasure memo. It recommends candidate FHWA Proven Safety Countermeasures, each with a CMF Clearinghouse identifier, and gives a benefit/cost estimate with its assumptions. Every number in the memo came from a tool result. The engineer edits the text, approves it, and exports it as a PDF for a project scoping meeting.

4.3. Proving the Ranking Against the Consultant HIN

Before the City Council presentation, the team opens the backtest page. It shows, for the HIN, the negative binomial EB model, and the ATSC model, the share of 2023–2025 F&SI crashes that occurred on each method's flagged sites, with 95% confidence intervals and the results at other network shares. If the ATSC model does not beat the HIN, that finding is reported as a valid result, and the project's value rests on the countermeasure memos and their cost.

4.4. Handling Failures and Requests Outside Scope

During a review meeting the TAMU AI Chat API becomes unavailable. The map, site pages, and backtest stay up, and new memo requests are marked "pending" until the service returns. In a separate case, a generated memo contains a crash count that does not appear in its tool results; the grounding check blocks the memo, logs the reason, and the memo is regenerated. When a reviewer asks the agent to add design-level detail to an intersection memo, such as signal timing or lane widths, the memo stays at planning level and states that an engineering study by a qualified engineer is required.

5. Analysis

5.1. Summary of Proposed Improvements

- A ranking that corrects for regression to the mean, so funds go to sites with persistent risk rather than sites that had a few unlucky years.
- Measured evidence: the ranking is scored on crashes that occurred after its data window, side by side with the consultant's HIN.
- Transparent reasoning: every flagged site shows its top risk factors, and every memo cites its sources.
- Faster and cheaper first drafts of countermeasure memos, at a logged cost per memo.
- A repeatable in-house process the City can rerun each year when new crash data is published.

5.2. Disadvantages and Limitations

- Crash coordinates and severity codes are noisy; map-matching and severity weighting reduce but do not remove this error.
- The model may not beat the HIN. The plan treats that outcome as a reportable result, not a failure.
- About 100 F&SI crashes occur per year in Brazos County, so the three-year test window limits how small a difference between methods can be detected.
- LLM drafts can still contain errors in wording or judgment; grounding checks block invented numbers and citations, and human review remains required.
- The ATSC reproduces the prioritization and memo portions of a safety action plan, not its public engagement or engineering design work.
- Operation depends on the TAMU AI Chat API during the project and on a City-supplied endpoint afterward.

5.3. Alternatives

- **Continue using outside consultants:** established expertise and accountability, but recurring cost and no internal capacity.
- **Use the Highway Safety Manual method alone:** a negative binomial model with EB is the industry standard and needs no machine learning. It is included as method B, so the backtest shows whether the ML model adds anything beyond EB.
- **Buy a commercial safety-analytics product:** quick to adopt, but adds licensing cost and is not tested against the City's own consultant deliverable.
- **Have staff use a general-purpose chatbot:** low cost, but no access to crash data, no computed numbers, no citations, and no measure of quality.
- **Keep the Revision B general consulting assistant:** broader reach, but a small, partly unobtainable evaluation set and no objective score. Not selected.

5.4. Impact

- **Financial:** a comparable consultant safety action plan costs about $200,000. The ATSC targets $1.00 or less in API charges per memo, plus student time and a low-cost host. Savings are reported only against the portions of the plan the system reproduces, so they are not overstated.

- **Societal:** better-targeted safety projects support the region's Vision Zero goal of eliminating traffic deaths and serious injuries by 2035. Reduced purchasing could affect local consulting firms.
- **Ethical:** AI output could influence public spending, so human review, source citations, and logging are required, consistent with the NIST AI Risk Management Framework. The system processes only public data and makes no statements about individuals; public use of its outputs should disclose that AI was involved.
- **Environmental:** the risk model trains on a CPU, the system runs on a low-power host, and LLM use is limited to one memo per flagged site and cached where possible.

5.5. Risk Register

Table 9 lists the main project risks, their likelihood and impact, the mitigation, and the owner. The full register with triggers is in the Execution and Validation Plan.

Table 9. Risk Register

Risk	Likelihood	Impact	Mitigation	Owner
Model does not beat the HIN	Medium	Medium	Backtest first (M2); report EB-versus-HIN as a valid result	Param Shelar (S2)
MPO HIN GIS layer not provided	Medium	High	Request in week 1; escalate through the City; reconstruct from published maps	Param Shelar (lead)
Crash location noise	High	Medium	Generous, logged match tolerance; report match rate	Kabir Singh (S1)
Data leakage invalidates backtest	Low	High	As-of dates on every feature (S1); automated cutoff check before every backtest	Param Shelar (S2)
LLM invents numbers or citations	Medium	High	Tools compute all numbers; grounding check blocks memo; human review	Jose Ruiz Garcia (S3)
Host too small for the full stack	Low	Medium	Container resource limits; CPU-only model; test on the Raspberry Pi 5 early	Juan Lopez (S4)
Reviewer not available for blind grading	Medium	Medium	Recruit through TTI by January; faculty reviewer as fallback	Juan Lopez (liaison)
Schedule slip; competing team on the same mandate	High	Medium	Narrow scope; weekly task review; named owner per milestone	Param Shelar (lead)

AI TRAFFIC-SAFETY CONSULTANT

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FUNCTIONAL SYSTEM REQUIREMENTS
AI CORPS – TEAM 4

REVISION – Draft
30 September 2026

FUNCTIONAL SYSTEM REQUIREMENTS
FOR
AI Traffic-Safety Consultant

PREPARED BY:

AI Corps #4 Oct 1, 2026

Author Date

APPROVED BY:

Param Shelar Oct 1, 2026

Project Leader Date

John Lusher, P.E. Date

T/A Date

Change Record

Table 1 List of changes

Rev.	Date	Originator	Approvals	Description
-	9/30/2026	Juan Lopez, Jose Ruiz Garcia, Param Shelar, Kabir Singh		Draft Release
-	10/1/2026	Param Shelar		Aligned with ConOps Rev C: four subsystems with named owners and requirement allocation (Table 4); editorial corrections
-	10/1/2026	Param Shelar		Scope reduction: chat page moved to a stretch goal, NVIDIA Jetson dropped, memo export PDF only, fixed intersection rule (three or more legs, 250 ft zone)

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1. Introduction

1.1. Purpose and Scope

This specification defines the technical requirements for the AI Traffic-Safety Consultant (ATSC), a software system developed by AI Corps Team 4 for the City of College Station. The ATSC estimates where fatal and serious-injury crashes are most likely to occur on College Station roads, tests those estimates against the High-Injury Network (HIN) produced for the region by an outside consultant, and drafts the countermeasure recommendations a traffic-safety consultant would deliver for each high-risk site.

The system has three deliverable parts: (1) a crash-risk model that produces an expected fatal and serious-injury crash rate for every road segment and intersection; (2) a countermeasure agent that uses a large language model (LLM) to write cited project memos whose numbers are all computed by deterministic tools; and (3) a web application that presents the risk map, site details, memos, and a backtest page comparing the model with the consultant HIN on crashes that occurred after the consultant's data window ended. Figure 1 shows a potential user view.

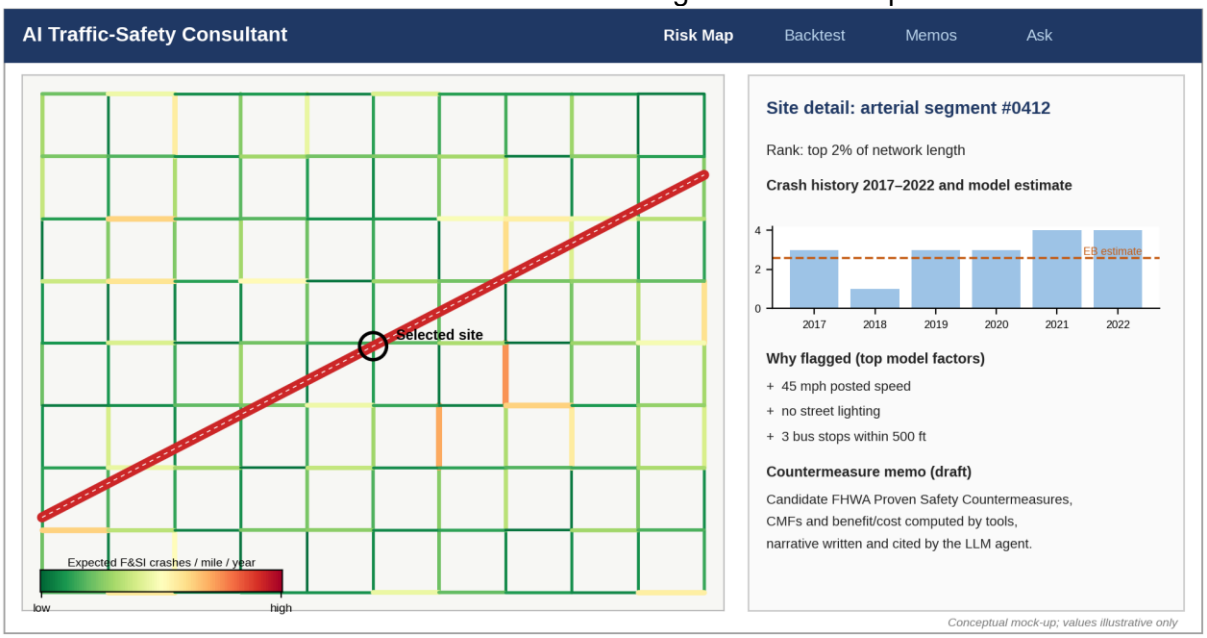


Figure 1. AI Traffic-Safety Consultant Conceptual Image

The project answers one question: can an LLM-based tool reproduce a paid consulting product at comparable quality and substantially lower cost? The reference deliverable is the High-Injury Network in the Bryan/College Station MPO Comprehensive Safety Action Plan, prepared by Kimley-Horn from 2017–2022 crash history. The operational concept is defined in the Team 4 Concept of Operations, Revision C, and the verification of each requirement is defined in the separate Execution and Validation Plan.

The following definitions differentiate between requirements and other statements.

Shall: This is the only verb used for the binding requirements.

Should/May: These verbs are used for stating non-mandatory goals.

Will: This verb is used for stating facts or declaration of purpose.

1.2. Responsibility and Change Authority

The team leader (Param Shelar) is responsible for ensuring that the ATSC meets the requirements in this document. Each subsystem owner is responsible for the requirements allocated to that subsystem in Table 4: S1 Road Network, Features, and Database (Kabir Singh); S2 Risk Model and Backtest (Param Shelar); S3 Countermeasure Agent (Jose Ruiz Garcia); and S4 Web Application and Deployment (Juan Lopez). Requirement changes require approval from the team leader, the program sponsor (Professor Nowka), and the City of College Station (point of contact: Sam Rivera). Changes are proposed at the team's biweekly meeting, reviewed with the course T/A, recorded in the Change Record, and released as a new revision of this document.

2. Applicable and Reference Documents

2.1. Applicable Documents

The following documents, of the exact issue and revision shown, form a part of this specification to the extent specified herein:

Table 2. Applicable Documents

Document Number	Revision/Release Date	Document Title
AASHTO HSM-1	1st Edition, 2010 (with 2014 Supplement)	Highway Safety Manual. Part B (network screening, Empirical Bayes method) and Part D (crash modification factors).
FHWA PSC	2021 update	FHWA Proven Safety Countermeasures
CMF Clearinghouse	Online database, accessed 2026	FHWA Crash Modification Factors Clearinghouse
TxDOT CRIS	Current extract, 2015–2025 data	Texas Crash Records Information System data and field definitions
WCAG 2.1	W3C Recommendation, June 2018	Web Content Accessibility Guidelines, Level AA
ATSC-ICD	Revision – (Draft), 9/30/2026	AI Traffic-Safety Consultant Interface Control Document
ATSC-EVP	Revision – (Draft), 9/30/2026	AI Traffic-Safety Consultant Execution and Validation Plan

2.2. Reference Documents

The following documents are reference documents utilized in the development of this specification. These documents do not form a part of this specification and are not controlled by their reference herein.

Table 3. Reference Documents

Document Number	Revision/Release Date	Document Title
BCS MPO CSAP	Study began Oct. 2023	Bryan/College Station MPO Comprehensive Safety Action Plan (Kimley-Horn), including the High-Injury Network
MPO Safety Briefing	January 2026	MPO safety presentation: Brazos County crash statistics and Safe System approach
Team 4 CONOPS	Revision C, 10/1/2026	AI Traffic-Safety Consultant Concept of Operations (AI Corps Team 4)
Persaud et al.	Accident Analysis & Prevention	Regression-to-the-mean and Empirical Bayes correction in hotspot identification
CGAN-EB	2022, Int. J. Transp. Sci. Technol.	Conditional GAN-based Empirical Bayes for crash prediction and hotspot identification
TTI Tip Card	2022	TTI Crash Location Coordinates Tip Card (crash coordinate accuracy)
arXiv:2306.05685	2023	Zheng et al., Judging LLM-as-a-Judge (LLM evaluation practice)

2.3. Order of Precedence

In the event of a conflict between the text of this specification and an applicable document cited herein, the text of this specification takes precedence without any exceptions.

All specifications, standards, exhibits, drawings or other documents that are invoked as "applicable" in this specification are incorporated as cited. All documents that are referred to within an applicable report are considered to be for guidance and information only, except ICDs that have their relevant documents considered to be incorporated as cited.

3. Requirements

This section defines the minimum requirements that the AI consultant must meet. The consultant is a software system; it delivers no custom hardware. Template sections that normally cover hardware (physical, electrical, environmental) are therefore written for the hosting platform, data interfaces, and operating environment of the software.

3.1. System Definition

Brazos County recorded more than 20,000 crashes from 2017 to 2022, including 101 fatal crashes and 524 serious-injury crashes, and the MPO, Brazos County, Bryan, and College Station have adopted Vision Zero goals for 2035. The consultant's High-Injury Network ranks roads by past crash counts, a method that suffers from regression to the mean: a road with a few unlucky years looks dangerous and then "improves" on its own. The ATSC replaces this raw-history ranking with a predictive model corrected by the Empirical Bayes (EB) method, proves or disproves the improvement with a held-out backtest, and automates the memo writing a consultant performs for each flagged site. Figure 2 shows the four subsystems, their owners, and how they connect.

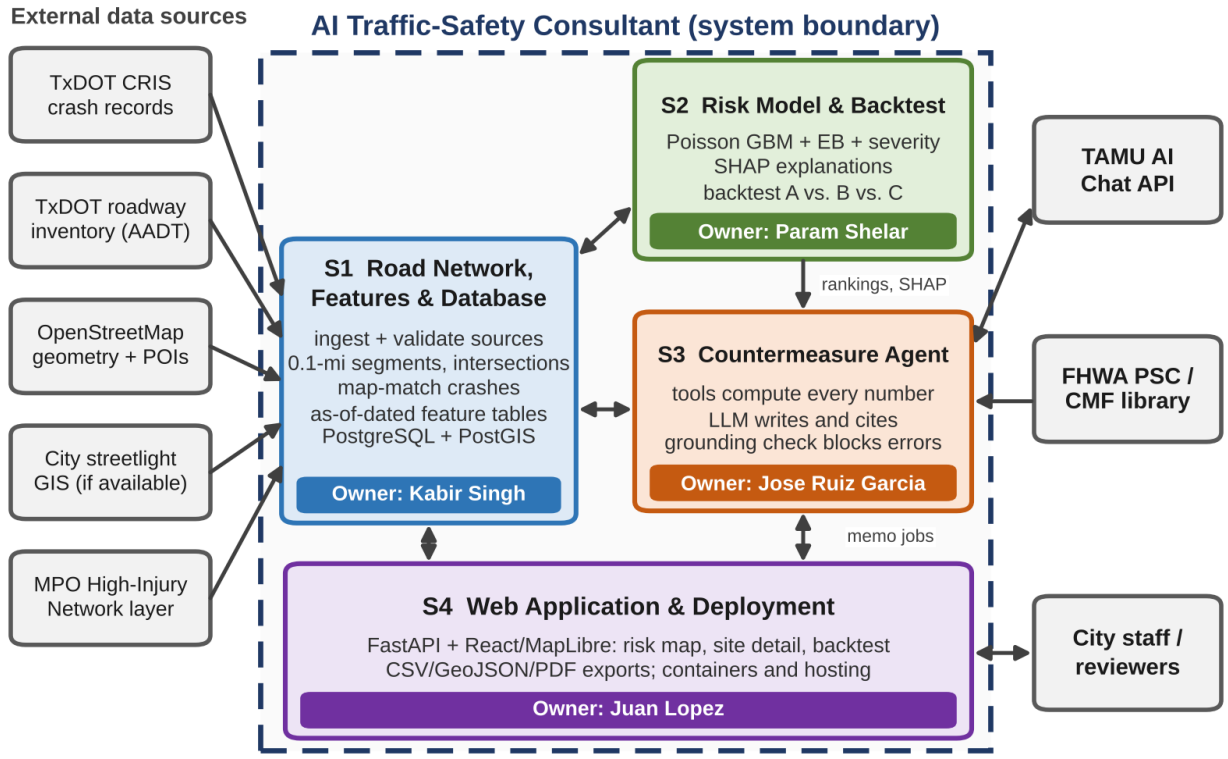


Figure 2. Block Diagram of System

S1 Road Network, Features, and Database (owner: Kabir Singh). Ingests TxDOT CRIS crash records (Brazos County, 2015–2025), the TxDOT roadway inventory (lanes, speed limit, AADT), OpenStreetMap geometry and points of interest, the city streetlight GIS layer if available, and the MPO HIN layer. It validates each source, splits the network into 0.1-mile segments and intersections, map-matches each crash to a site, computes features with as-of dates, and stores all data, model runs, predictions, and memos in a PostgreSQL/PostGIS database shared by the other subsystems.

S2 Risk Model and Backtest (owner: Param Shelar). Fits a gradient-boosted model with Poisson loss, estimates overdispersion, blends predictions with each site's own history using EB weights, applies a severity model to obtain expected F&SI crashes per mile per year, and ranks sites. SHAP values explain each prediction. The backtest compares three methods, (A) the consultant HIN, (B) a negative binomial safety performance function with EB, and (C) the ATSC ML-EB model, on 2023–2025 crashes.

S3 Countermeasure Agent (owner: Jose Ruiz Garcia). For each flagged site, calls tools that return the crash profile, matching FHWA Proven Safety Countermeasures and crash modification factors (CMFs), and a benefit/cost estimate. The LLM, accessed through the TAMU AI Chat API, writes and cites the memo but does not originate any number. If the stretch-goal chat page is delivered, S3 also answers its questions with citations to the records used.

S4 Web Application and Deployment (owner: Juan Lopez). A FastAPI backend and React/MapLibre frontend with three core pages: risk map, site detail (history, prediction, memo), and backtest results, plus CSV, GeoJSON, and PDF exports. A question-answering chat page is a stretch goal. S4 also owns the container images and deployment on the host.

Table 4 allocates every requirement in Section 3.2 to the subsystem whose owner is responsible for meeting and verifying it. Requirements marked "All" are met by every subsystem for its own part and are integrated and verified by the team leader.

Table 4. Subsystem Ownership and Requirement Allocation

Subsystem	Owner	Allocated requirements (Section 3.2)
S1 Road Network, Features, and Database	Kabir Singh	3.2.1.2 Site Definition; 3.2.1.3 Crash Data Ingest; 3.2.1.4 Map-Matching Rate; 3.2.1.5 Temporal Leakage Control; 3.2.3.1.1 Input Data Sources; 3.2.3.1.2 Input Validation; 3.2.4.4 Data Classification
S2 Risk Model and Backtest	Param Shelar	3.2.1.1 Network Coverage; 3.2.1.6 Comparison Methods; 3.2.1.7 Empirical Bayes Correction; 3.2.1.8 Severity Estimate; 3.2.1.9 High-Risk Flagging; 3.2.1.10 Backtest Reporting; 3.2.1.11 Predictive Performance Goal; 3.2.1.12 Prediction Explanations; 3.2.1.21 Reproducibility; 3.2.2.2 Model Training Time; 3.2.3.2.3 Model Artifact Output
S3 Countermeasure Agent	Jose Ruiz Garcia	3.2.1.13 Countermeasure Memo Content; 3.2.1.14 Numeric Grounding; 3.2.1.15 Citation Grounding; 3.2.1.16 Memo Quality Goal; 3.2.1.17 Cost and Time per Memo
S4 Web Application and Deployment	Juan Lopez	3.2.1.18 Web Application Functions; 3.2.1.19 Web Responsiveness; 3.2.1.20 Exports; 3.2.2.1 Hosting Footprint; 3.2.2.3 Deployment; 3.2.3.2.1 Data Output; 3.2.3.3 Network Interfaces; 3.2.4.1 Server Environment; 3.2.4.2 Client Environment; 3.2.4.3 Accessibility; 3.2.4.5 Credential Handling; 3.2.5.1.2 Isolation and Recovery
All (integrated by team leader)	Param Shelar	3.2.3.1 Inputs (a, b); 3.2.3.1.3 External Commands; 3.2.3.2.2 Diagnostic Output; 3.2.5 Failure Propagation; 3.2.5.1.1 Built-In Test and 3.2.5.1.1.1 to 3.2.5.1.1.3

3.2. Characteristics

3.2.1. Functional / Performance Requirements

3.2.1.1. Network Coverage

The AI consultant shall produce an expected fatal and serious-injury crash rate for every road segment and intersection in the analysis network, defined as public roads inside the City of College Station limits that appear in the TxDOT roadway inventory or OpenStreetMap.

Rationale: The consultant HIN covers the full network; a partial ranking cannot be compared to it.

3.2.1.2. Site Definition

The AI consultant shall divide roads into segments of 0.1 mile (±0.05 mile) and shall treat each network node where three or more public road legs meet as an intersection site. Each intersection site shall have an influence zone of 250 ft from its center; crashes inside the zone are assigned to the intersection, and crashes outside it to the nearest segment.

Rationale: Segment length is the offset in the risk model and the denominator of the ranking rate, so a consistent length keeps rates comparable across sites. A fixed intersection rule makes site assignment repeatable and auditable; 250 ft is a widely used planning-level intersection influence distance.

3.2.1.3. Crash Data Ingest

The AI consultant shall ingest TxDOT CRIS crash records for Brazos County for calendar years 2015 through 2025, retaining crash date, location, KABCO severity, crash type, and contributing roadway attributes as defined in the ICD.

Rationale: CRIS is the public statewide source with coordinates and bulk extracts; 2015–2022 provide the history used for training, and 2023–2025 provide the held-out test window.

3.2.1.4. Map-Matching Rate

The AI consultant shall assign at least 95% of geocoded fatal and serious-injury crashes in the analysis area to a segment or intersection and shall log every unmatched crash with the reason it was not matched.

Rationale: Crash coordinates are noisy (in one TTI sample only 25% of officer and TxDOT coordinates agreed within 50 ft), so match tolerance must be generous but auditable.

3.2.1.5. Temporal Leakage Control

For backtest runs, every model input shall be knowable as of 31 December 2022. The AI consultant shall record the as-of date of each feature source and shall refuse to run a backtest if any feature is dated after the cutoff.

Rationale: A model that sees post-2022 information would produce an invalid backtest and an unfair comparison with the HIN.

3.2.1.6. Comparison Methods

The AI consultant shall implement three ranking methods on identical inputs: (A) the consultant HIN as published by the MPO, (B) a negative binomial safety performance function with EB adjustment per the Highway Safety Manual, and (C) the AI consultant gradient-boosted model with EB blending (ML-EB).

Rationale: Method B is the industry standard; comparing against both A and B shows whether gains come from EB alone or from the machine-learning model.

3.2.1.7. Empirical Bayes Correction

The ML-EB method shall combine model-predicted crashes with observed crashes at each site using the weight $w = 1 / (1 + k \cdot \Sigma\mu)$, where k is the fitted overdispersion parameter and $\Sigma\mu$ is the predicted crash total over the history years.

Rationale: EB is the standard correction for regression to the mean; ranking on raw counts overstates sites with unlucky years.

3.2.1.8. Severity Estimate

The AI consultant shall estimate, for each site, the probability that a crash at that site is fatal or serious (KABCO K or A) and shall rank sites on expected fatal and serious-injury crashes per mile per year.

Rationale: Vision Zero targets deaths and serious injuries, not total crashes.

3.2.1.9. High-Risk Flagging

The AI consultant shall flag the highest-ranked sites whose combined length equals 11% (±0.5%) of total network length.

Rationale: The consultant HIN covers about 11% of roads; matching that share makes the capture rates directly comparable.

3.2.1.10. Backtest Reporting

The AI consultant shall compute, for each of methods A, B, and C, the capture rate C = (2023–2025 fatal and serious-injury crashes on flagged sites) / (all 2023–2025 fatal and serious-injury crashes in the analysis area), with a 95% bootstrap confidence interval from at least 1,000 resamples, and shall display all three results on the backtest page.

Rationale: The backtest is the proof. The team reports whatever it shows, including a result where the model does not beat the HIN.

3.2.1.11. Predictive Performance Goal

The ML-EB method should achieve a higher capture rate than the consultant HIN on the 2023–2025 test window, with the difference reported with its confidence interval.

Rationale: This is the objective, not the threshold. Stated as a goal because the outcome is unknown until the backtest runs; an EB-over-HIN result alone is still a defensible finding.

3.2.1.12. Prediction Explanations

For every flagged site, the AI consultant shall list the five features with the largest SHAP contributions to its prediction, in plain-language form (for example, "45 mph posted speed").

Rationale: City staff need to know why a site was flagged; the explanations also feed the countermeasure memo.

3.2.1.13. Countermeasure Memo Content

For each flagged site, the AI consultant shall generate a memo containing: the site's crash profile by type and severity; the model estimate and top factors; at least one applicable FHWA Proven Safety Countermeasure with its CMF and source; an estimated benefit/cost ratio with stated assumptions; and a reference list.

Rationale: These are the elements of a consultant project sheet for a safety action plan.

3.2.1.14. Numeric Grounding

Every number in a memo shall come from a tool result recorded with the memo. The AI consultant shall automatically check each generated memo and shall block publication of any memo that contains a number not present in its tool results.

Rationale: LLMs can invent figures; separating computation (tools) from writing (LLM) makes every number traceable.

3.2.1.15. Citation Grounding

Each countermeasure recommendation in a memo shall cite the CMF Clearinghouse identifier or FHWA source from which it was retrieved.

Rationale: Reviewers must be able to trace each recommendation to a recognized source.

3.2.1.16. Memo Quality Goal

In blind grading by a traffic-safety reviewer (a TTI safety researcher or equivalent) using the rubric in the Execution and Validation Plan, AI consultant memos should score a mean of at least 3.5 out of 5 and should not score lower than consultant-style reference memos by more than 0.5 point.

Rationale: Cost savings matter only if quality is comparable; the rubric makes "comparable" measurable.

3.2.1.17. Cost and Time per Memo

The AI consultant shall log LLM token usage and cost for every memo. Memo generation should take no more than 2 minutes and should cost no more than $1.00 in API charges per site.

Rationale: The project compares tool cost with consultant fees; API charges are covered by the program but require approval before exceeding budget.

3.2.1.18. Web Application Functions

The web application shall provide: (a) a risk map of all sites colored by expected fatal and serious-injury rate, with the flagged network highlighted; (b) a site detail page with crash history, model estimate, SHAP factors, and memo; and (c) a backtest page with capture rates and confidence intervals for methods A, B, and C. As a stretch goal, the web application should also provide (d) a chat page that answers questions using AI consultant data and cites the records it used.

Rationale: Views (a) to (c) are what City staff need to act on the ranking. The chat page is a stretch goal because it adds a second LLM feature with its own grounding risk and does not affect the core result.

3.2.1.19. Web Responsiveness

With up to five concurrent users on the campus network, the risk map shall display the full network within 5 seconds, and a site detail page shall load within 2 seconds.

Rationale: Expected use is a small group of city staff and reviewers; these limits keep the tool usable in a live meeting.

3.2.1.20. Exports

The AI consultant shall export the ranked site list (CSV and GeoJSON) and each memo (PDF).

Rationale: City staff work in GIS and office tools; exports let results enter existing workflows.

3.2.1.21. Reproducibility

Each model run shall be stored with a unique run identifier, input data versions, code version, random seed, and hyperparameters, and re-running with the same identifier shall reproduce the ranking exactly.

Rationale: Results presented to City Council must be reproducible and auditable.

3.2.2. Physical Characteristics

The AI consultant delivers no physical hardware. The requirements below cover the computing footprint the software needs.

3.2.2.1. Hosting Footprint

The complete AI consultant (database, backend, frontend, and batch jobs) shall run on a single Linux host with no more than 4 CPU cores, 8 GB of RAM, and 128 GB of storage, and shall not require a GPU. The allocation of these resources to containers is defined in the ICD.

Rationale: Matches the Raspberry Pi 5 the team will use as its reference test host and keeps hosting within what Texas A&M or the City can provide at little cost; gradient boosting on a county-scale network does not need a GPU.

3.2.2.2. Model Training Time

A full model training and backtest run shall complete within 4 hours on the host defined above.

Rationale: Allows a rerun within a working day after data or code changes.

3.2.2.3. Deployment

The AI consultant shall be packaged as container images with a single configuration file, and deployment details shall be captured in the AI consultant ICD.

Rationale: Lets the City or a future team redeploy the system without the original developers.

3.2.3. Electrical and Data Interface Characteristics

The AI consultant has no electrical interfaces of its own; power, voltage, ripple, connector, and wiring requirements in the template are not applicable. This section instead defines the software inputs and outputs. All interface details, from file formats to API message structure, are captured in the ATSC ICD.

3.2.3.1. Inputs

a. The presence, absence, or malformation of any input file or record, in any sequence, shall not corrupt stored data or published results; invalid records shall be rejected and logged.
b. No sequence of user actions or API requests shall corrupt stored data or crash the service.

Rationale: By design, limits the chance of corrupted results from operator or user error.

3.2.3.1.1 Input Data Sources

The AI consultant shall accept the data sources listed in the ICD (CRIS extract, TxDOT roadway inventory, OpenStreetMap extract, city streetlight GIS, MPO HIN layer, and the countermeasure library) in the formats defined there.

Rationale: Fixes the interface so each source can be refreshed independently.

3.2.3.1.2 Input Validation

On ingest, the AI consultant shall check record counts, required fields, coordinate bounds, date ranges, and severity codes, and shall stop the pipeline if more than 2% of records fail validation.

Rationale: A silent data problem would corrupt the ranking without any visible error.

3.2.3.1.3 External Commands

The AI consultant shall document all API endpoints and operator commands in the ICD.

Rationale: The ICD captures all interface details from file formats to request and response structure.

3.2.3.2. Outputs

3.2.3.2.1 Data Output

The AI consultant shall provide ranked results, memos, and backtest results through the REST API and export formats defined in the ICD.

Rationale: Results pass directly to city staff and reviewers.

3.2.3.2.2 Diagnostic Output

The AI consultant shall write structured logs for every pipeline stage, model run, LLM call (prompt, tool calls, tokens, cost, latency), and API request.

Rationale: Provides the ability to debug runs and audit how each memo was produced.

3.2.3.2.3 Model Artifact Output

The AI consultant shall export trained model files, feature tables, and per-site predictions for each run.

Rationale: Allows independent review of the model by TTI or city analysts.

3.2.3.3. Network Interfaces

All client and external API traffic shall use HTTPS with TLS 1.2 or later.

Rationale: Protects API keys and prevents tampering with published results.

3.2.4. Environmental Requirements

The AI consultant shall operate in the software environments specified in the following sections.

Rationale: For a software system, the operating environment replaces the thermal, altitude, and contamination environments in the template.

3.2.4.1. Server Environment

The server components shall run on a current long-term-support Linux distribution with Docker, Python 3.11 or later, and PostgreSQL 15 or later with PostGIS 3.

Rationale: Widely supported open-source stack with no license cost to the City.

3.2.4.2. Client Environment

The web application shall work in current versions of Chrome, Edge, Firefox, and Safari on desktop and tablet screens of 1280 × 800 pixels or larger, with no plug-ins.

Rationale: City staff use standard browsers; no installation should be required.

3.2.4.3. Accessibility

The web application should conform to WCAG 2.1 Level AA, including a non-color cue for risk levels on the map.

Rationale: Public-agency web tools are expected to meet accessibility guidelines; the City has an active digital-accessibility solicitation (RFP 26-084).

3.2.4.4. Data Classification

The AI consultant shall process only public data and shall not store personal identifiers from crash records beyond what the public CRIS extract contains.

Rationale: The TAMU AI Chat API is approved for data up to University-Confidential and does not train external models on inputs, but public-only data keeps the system simple to share with the City.

3.2.4.5. Credential Handling

API keys shall be stored in environment variables or a secrets file excluded from version control, never in source code or the browser.

Rationale: Prevents leaked keys and unapproved API charges.

3.2.5. Failure Propagation

A fault in one AI consultant subsystem shall not propagate to published results or to other subsystems.

3.2.5.1. Failure Detection, Isolation, and Recovery (FDIR)

3.2.5.1.1 Built-In Test (BIT)

The AI consultant shall include an automated test suite that runs on every code change and before every model run, checking data validation, model outputs (non-negative rates, weights between 0 and 1, rank stability), memo grounding, and API health.

3.2.5.1.1.1 BIT Critical Fault Detection

The BIT shall detect at least 95% of seeded critical faults (for example, a shifted date column, swapped coordinates, a post-cutoff feature, or an invented number in a memo) in the fault-injection test set defined in the Execution and Validation Plan.

Rationale: These faults would silently invalidate the backtest or mislead city staff.

3.2.5.1.1.2 BIT False Alarms

The BIT shall have a false alarm rate of less than 5% on known-good runs.

Rationale: Frequent false alarms train operators to ignore the checks.

3.2.5.1.1.3 BIT Log

The BIT shall save the result of each test run to a log retrievable from the operator interface.

Rationale: Gives maintainers a record of when and why a run was blocked.

3.2.5.1.2 Isolation and Recovery

If the LLM API is unavailable, the risk map, site details, and backtest shall remain available and affected memos shall be marked "pending." A failed model run shall leave the last published run in place, and operators shall be able to roll back to any stored run.

Rationale: The risk ranking is useful on its own and should not depend on an external service.

4. Support Requirements

The team will deliver the following with the AI consultant at project completion (spring 2027):

- Source code repository with build and test instructions
- Container images and a deployment guide (see ICD)
- User guide for city staff covering the map, site pages, memos, and exports
- Data dictionary and a model card describing training data, methods, backtest results, and known limitations
- Operator runbook for the annual data refresh (new CRIS year) and model re-run
- Evaluation package: backtest results, grading rubric, and blind-grading results
- Background research completed, linked through the GitHub repository

The City must provide, or arrange through Texas A&M, a host meeting Section 3.2.2 and an API key for the TAMU AI Chat API or an equivalent LLM service. The AI consultant is a decision-support tool: recommendations are drafts for review by qualified traffic engineers, and the system is not a substitute for a PE-sealed engineering study. Team support ends at the close of the capstone course unless the City and the program agree otherwise; issues found before then are tracked in the repository issue tracker and addressed at the biweekly team meeting.

Appendix A: Acronyms and Abbreviations

AADT	Annual Average Daily Traffic
API	Application Programming Interface
B/C	Benefit/Cost ratio
BIT	Built-In Test
CI	Confidence Interval
CMF	Crash Modification Factor
CRIS	Crash Records Information System (TxDOT)
CSAP	Comprehensive Safety Action Plan
CSV	Comma-Separated Values
EB	Empirical Bayes
FDIR	Failure Detection, Isolation, and Recovery
FHWA	Federal Highway Administration
GBM	Gradient-Boosted Model
GIS	Geographic Information System
HIN	High-Injury Network
HSM	Highway Safety Manual (AASHTO)
HTTPS	Hypertext Transfer Protocol Secure
ICD	Interface Control Document
KABCO	Injury severity scale: K fatal, A suspected serious, B minor, C possible, O none
LLM	Large Language Model
ML	Machine Learning
MPO	Metropolitan Planning Organization (Bryan/College Station)
NB	Negative Binomial
OSM	OpenStreetMap
PSC	Proven Safety Countermeasure (FHWA)
SHAP	SHapley Additive exPlanations
SPF	Safety Performance Function
TBD	To Be Determined
TLS	Transport Layer Security
TTI	Texas A&M Transportation Institute
TxDOT	Texas Department of Transportation
WCAG	Web Content Accessibility Guidelines

Appendix B: Definition of Terms

Backtest	Ranking sites with data through 2022, then scoring the ranking on crashes that occurred in 2023–2025.
Capture rate	Share of test-window F&SI crashes that occurred on flagged sites.
Empirical Bayes	A weighted average of a site's predicted and observed crashes that corrects for regression to the mean.
Flagged network	The highest-ranked sites whose total length is 11% of the network.
High-Injury Network	The consultant's set of roads carrying a disproportionate share of F&SI crashes.
Leakage	Use of information in model training that would not have been available at the prediction date.
Overdispersion	Crash-count variance larger than the mean, modeled as $\text{Var} = \mu + k\mu^2$.
Regression to the mean	The tendency of sites with unusually high counts in one period to show lower counts in the next by chance.
Site	A road segment of about 0.1 mile or an intersection.
Tool (agent)	A deterministic function the LLM calls to retrieve or compute data; the LLM itself computes no numbers.

Appendix C: Interface Control Documents

All ATSC interfaces are defined in the AI Traffic-Safety Consultant Interface Control Document (ATSC-ICD), Revision – (Draft), dated 30 September 2026, which is incorporated into this specification by reference.

AI TRAFFIC-SAFETY CONSULTANT

Juan Lopez
Jose Ruiz Garcia
Param Shelar
Kabir Singh

INTERFACE CONTROL DOCUMENT
AI CORPS – TEAM 4

REVISION – Draft
30 September 2026

INTERFACE CONTROL DOCUMENT
FOR
AI Traffic-Safety Consultant

PREPARED BY:

_____
Author Kabir Singh Date Oct 1 2026

APPROVED BY:

Param Shelar Oct 1, 2026
_____
Project Leader Date

_____
John Lusher II, P.E. Date

_____
T/A Date

Change Record

Table 1 List of changes

Rev.	Date	Originator	Approvals	Description
-	9/30/2026	Kabir Singh		Draft Release
-	10/1/2026	Juan Lopez		Draft Final Edits
-	10/1/2026	Param Shelar		Aligned with ConOps Rev C and FSR: four subsystems with owners (database in S1), FSR hosting footprint in Table 2, updated Figures 1 and 2
-	10/1/2026	Param Shelar		Scope reduction: chat page moved to a stretch goal, NVIDIA Jetson dropped, memo export PDF only, fixed intersection rule (three or more legs, 250 ft zone)

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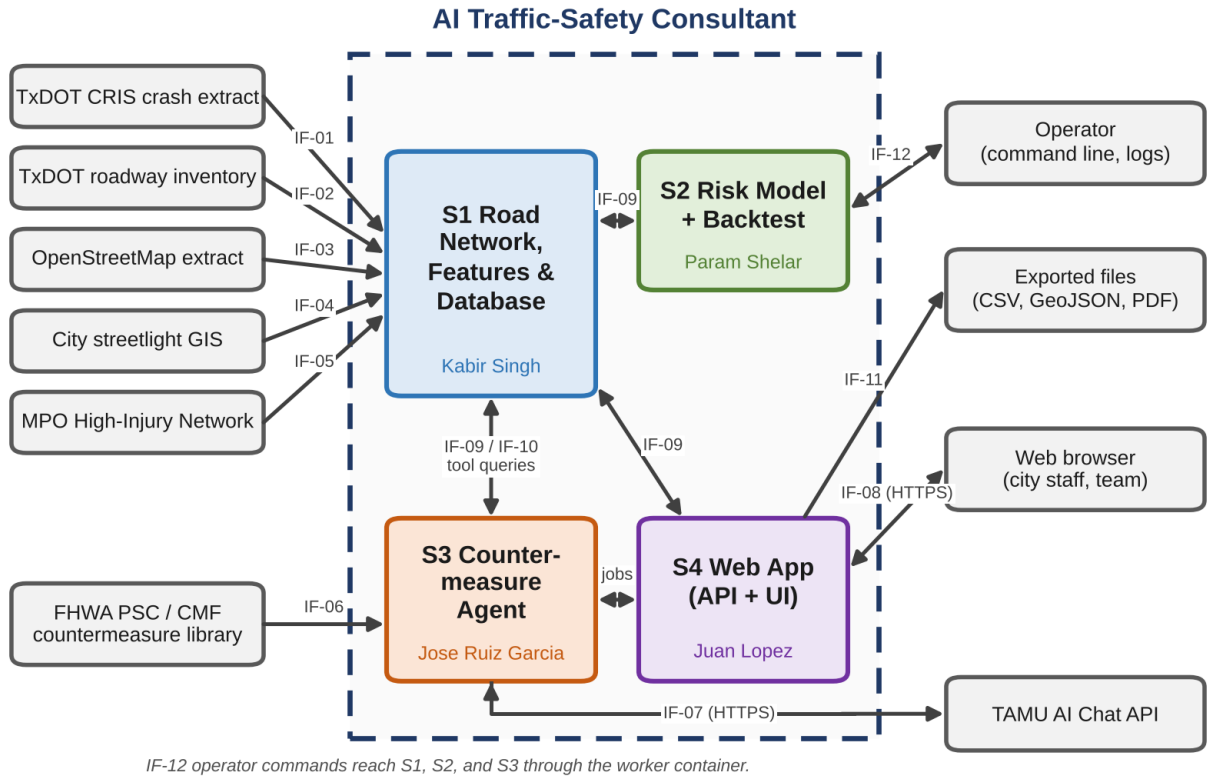
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1. Overview

This Interface Control Document (ICD) defines every interface of the AI Traffic-Safety Consultant (ATSC): the public data sources it ingests, the external LLM service it calls, the web API and user interface it presents, the files it exports, the operator commands that run it, and the internal interfaces between its four subsystems: S1 Road Network, Features, and Database (owner: Kabir Singh); S2 Risk Model and Backtest (owner: Param Shelar); S3 Countermeasure Agent (owner: Jose Ruiz Garcia); and S4 Web Application and Deployment (owner: Juan Lopez). It is the controlling reference for the interface requirements in the ATSC Functional System Requirements (FSR).

The ATSC is a software system. It has no custom hardware, so the physical, thermal, and electrical sections of this template describe the hosting platform, and the communications section carries most of the interface detail. Each interface has an identifier (IF-01 to IF-12) shown in Figure 1 and Table 3.



IF-12 operator commands reach S1, S2, and S3 through the worker container.

Figure 1. ATSC Interface Context Diagram

2. References and Definitions

The standards and data documents below govern the ATSC interfaces. Where a revision is listed, that revision applies.

2.1. References

ATSC Functional System Requirements
AI Traffic-Safety Consultant FSR
Revision – (Draft), 30 Sep 2026
RFC 9110
HTTP Semantics (IETF)
June 2022
RFC 8446 / RFC 5246
Transport Layer Security 1.3 / 1.2 (IETF)
Aug 2018 / Aug 2008
RFC 7946
The GeoJSON Format (IETF)
Aug 2016
RFC 8259
The JavaScript Object Notation (JSON) Data Interchange Format (IETF)
Dec 2017
RFC 4180
Common Format and MIME Type for Comma-Separated Values (CSV) Files (IETF)
Oct 2005
OpenAPI Specification 3.1
OpenAPI Initiative
2021
ISO 8601
Date and time format
Current edition
EPSG:4326 / EPSG:32614
WGS 84 geographic coordinates / WGS 84 UTM Zone 14N (meters)
EPSG registry
TxDOT CRIS
Crash Records Information System extracts and data dictionary
Current edition
AASHTO Highway Safety Manual
KABCO severity scale; Empirical Bayes method
1st Edition, 2010
FHWA CMF Clearinghouse
Crash Modification Factors Clearinghouse (IDs, star ratings)
Online, accessed 2026
WCAG 2.1
Web Content Accessibility Guidelines (W3C)
June 2018
TAMU AI Chat API documentation
docs.tamus.ai
Current, accessed 2026

2.2. Definitions

API	Application Programming Interface
ATSC	AI Traffic-Safety Consultant
CMF	Crash Modification Factor
CRIS	Crash Records Information System (TxDOT)
EB	Empirical Bayes
F&SI	Fatal and Serious Injury (KABCO K and A)
GeoJSON	JSON format for geographic features (RFC 7946)
HIN	High-Injury Network
HTTPS	HTTP over TLS
IF-nn	ATSC interface identifier
KABCO	Severity scale: K fatal, A suspected serious injury, B suspected minor, C possible, O no injury
LLM	Large Language Model
OSM	OpenStreetMap
PSC	FHWA Proven Safety Countermeasure
REST	Representational State Transfer (resource-oriented HTTP API)
Site	A road segment of about 0.1 mile or an intersection
TBD	To Be Determined
TLS	Transport Layer Security
UTC	Coordinated Universal Time

3. Physical Interface

The ATSC has no physical hardware of its own and no weight or mounting constraints. Its physical interface is the host it is deployed on. Figure 2 shows the deployment: five containers on one Linux host, reached by browsers over HTTPS, with outbound calls to the TAMU AI Chat API.

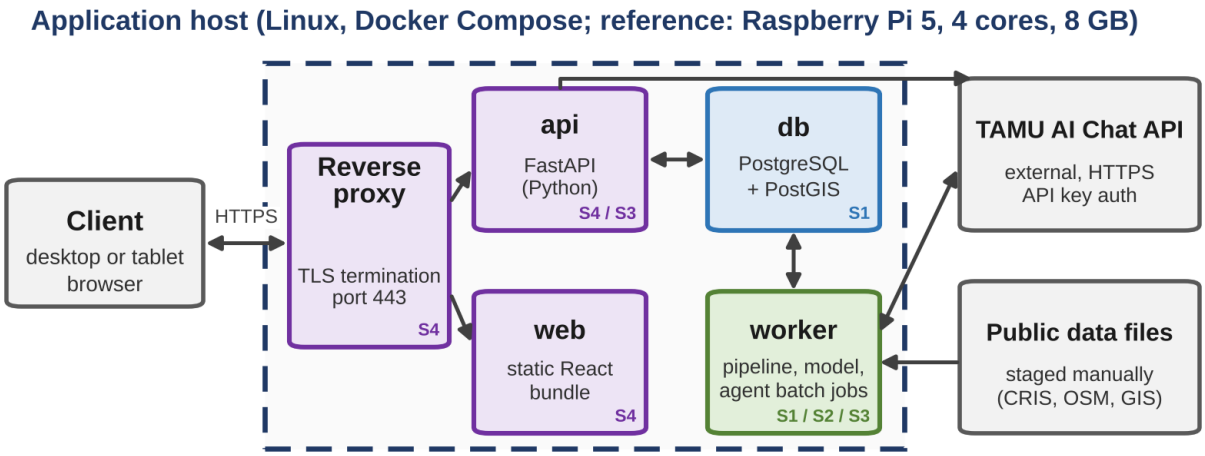


Figure 2. ATSC Deployment Diagram

3.1. Weight

Not applicable (software system).

3.2. Dimensions

The "dimensions" of the ATSC are the compute resources each container needs. The total must fit the hosting footprint in the FSR (4 CPU cores, 8 GB RAM, 128 GB storage, no GPU). The limits in Table 2 are Docker resource caps; batch jobs in the worker run outside interactive review sessions.

Table 2. Container Resource Allocation

Container	Subsystem	vCPU	RAM	Storage	Notes
proxy	S4	0.25	0.25 GB	1 GB	TLS termination, static caching
web	S4	0.25	0.25 GB	1 GB	Built React bundle served as static files
api	S4 / S3	0.5	1 GB	2 GB	FastAPI; on-demand memo requests (and chat, if delivered)
worker	S1, S2, S3	2	3.5 GB	20 GB scratch	Batch jobs: ingest, map-matching, training, backtest, memo batches
db	S1	1	2 GB	80 GB	PostgreSQL + PostGIS; includes run history and logs
Total		**4**	**7 GB**	**104 GB**	Fits the FSR footprint; leaves 1 GB RAM for the host OS and 24 GB storage headroom

3.3. Mounting Locations

The ATSC will be deployed on one of the following hosts, to be decided with the City and the program by the end of the fall semester:

- A Linux virtual machine provided by Texas A&M for the capstone (development and pilot)

- A Linux virtual machine provided by the City of College Station (production, if the City adopts the tool)
- Test platform: a Raspberry Pi 5 (4 cores, 8 GB, ARM64) as the reference host for the FSR hosting footprint; container images are built for both linux/amd64 and linux/arm64

Persistent data lives in two Docker volumes: the database volume and a data volume holding the raw source files staged for ingest. Both are backed up nightly to storage on the same host and weekly to a second location chosen by the host owner.

4. Thermal Interface

The ATSC has no thermal interface. It runs on a standard server or virtual machine in a climate-controlled data center, and no cooling, heatsink, or cold-wall provisions are required.

5. Electrical and Data Interface

The ATSC draws no power directly because the host provides power. Sections 5.1 and 5.2 are therefore not applicable. Section 5.3 defines the data interfaces that take the place of signal interfaces, and Sections 5.4 and 5.5 define the display and user control interfaces.

5.1. Primary Input Power

Not applicable. Power is supplied by the host platform.

5.2. Polarity Reversal

Not applicable.

5.3. Signal Interfaces (Data Interfaces)

Table 3 lists every ATSC interface. Direction is relative to the ATSC. The owner is the subsystem responsible for implementing and testing the interface: S1 Kabir Singh, S2 Param Shelar, S3 Jose Ruiz Garcia, and S4 Juan Lopez.

Table 3. Interface Summary

ID	Interface	Direction	Format / Protocol	Frequency	Owner
IF-01	TxDOT CRIS crash extract (Brazos County, 2015–2025)	In	CSV files, manual download	Once, then yearly	S1
IF-02	TxDOT roadway inventory (lanes, speed, AADT)	In	Shapefile or GeoJSON	Once, then yearly	S1
IF-03	OpenStreetMap extract (geometry, POIs)	In	.osm.pbf	Once per model run	S1
IF-04	City streetlight GIS (if available)	In	Shapefile or GeoJSON	Once	S1
IF-05	MPO High-Injury Network layer	In	Shapefile or GeoJSON	Once	S1 / S2
IF-06	Countermeasure library (FHWA PSC, CMF Clearinghouse)	In	Curated CSV	Per library release	S3
IF-07	TAMU AI Chat API	Out / In	HTTPS, JSON	Per memo or chat request	S3
IF-08	ATSC REST API to web browser	In / Out	HTTPS, JSON, GeoJSON	On demand	S4
IF-09	Database interface (internal, part of S1)	Internal	SQL over TCP 5432	Continuous	S1
IF-10	Agent tool interface (internal)	Internal	JSON function calls	Per memo	S3
IF-11	Exports	Out	CSV, GeoJSON, PDF	On demand	S4
IF-12	Operator command-line interface and logs	In / Out	Shell commands, JSON logs	Per run	S1–S3

5.3.1. IF-01 TxDOT CRIS Crash Extract

Crash records are exported from the public CRIS query tool for Brazos County, one CSV per year, and staged in the data volume. S1 maps the CRIS columns to the internal crash schema below; the mapping from CRIS column names is kept in a versioned field-map file so a CRIS format change touches only that file.

Table 4. Internal Crash Record Schema (IF-01)

Internal field	Type	Description	Validation
crash_id	text	CRIS crash identifier	Unique, not null
crash_date	date	Crash date (ISO 8601)	2015-01-01 to 2025-12-31
crash_time	time	Crash time, local	Optional
severity	char(1)	Highest injury severity in crash, KABCO (K, A, B, C, O)	In set; unknown mapped to "U" and excluded from F&SI
lat, lon	float	Crash coordinates, EPSG:4326	Inside Brazos County bounding box; null allowed, logged
crash_type	text	Manner of collision (angle, rear end, pedestrian, etc.)	Mapped to ATSC type list
light_cond	text	Light condition (daylight, dark-lit, dark-unlit, etc.)	Mapped to ATSC list
road_part / int_related	text / bool	Intersection-related flag and roadway part	Used by map-matching
ped_bike	bool	Pedestrian or bicyclist involved	Derived from unit types

5.3.2. IF-02 to IF-05 Geographic Inputs

Table 5. Geographic Input Requirements

ID	Required attributes	Coordinate system	Use
IF-02	Route ID, lanes, posted speed, functional class, median type, AADT and AADT year	Reprojected to EPSG:4326	Exposure and design features; AADT must be dated 2022 or earlier for backtest runs
IF-03	Road centerlines, intersections, sidewalks, bus stops, schools, bars and restaurants	EPSG:4326	Network geometry and pedestrian-context features; OSM snapshot date recorded as the feature as-of date
IF-04	Streetlight point locations	Any; reprojected	Lighting feature (lights per segment); if unavailable, OSM lit tags are used and the gap is logged
IF-05	HIN segment geometry	Any; reprojected	Method A in the backtest; matched to ATSC sites by overlap of 50% or more of segment length

All lengths and distances are computed in EPSG:32614 (UTM Zone 14N, meters) and reported in miles and feet.

5.3.3. IF-06 Countermeasure Library

The countermeasure library is a curated CSV maintained by the team from the FHWA Proven Safety Countermeasures list and the CMF Clearinghouse. Each row is one countermeasure and CMF pairing.

Table 6. Countermeasure Library Schema (IF-06)

Field	Type	Description
cm_id	text	ATSC countermeasure identifier
psc_name	text	FHWA Proven Safety Countermeasure name

Field	Type	Description
cmf_id	text	CMF Clearinghouse identifier
cmf	float	Crash modification factor (0 < CMF ≤ 1 for reductions)
crash_types	text list	Crash types the CMF applies to
severity_scope	text	Severities the CMF applies to (e.g., K,A or all)
site_type	text	Segment, signalized intersection, unsignalized intersection, or any
star_rating	int	CMF Clearinghouse quality rating (1–5)
unit_cost, service_life	float, int	Planning-level cost (USD) and service life (years), with source
source_url	text	Link to the FHWA or CMF Clearinghouse page

5.4. Video Interfaces (Display Interface)

The ATSC has no video hardware interface. Its display interface is the web map: sites are drawn as lines (segments) and circles (intersections) colored on a sequential scale of expected F&SI crashes per mile per year, with the flagged network drawn with a heavier line weight so risk is not conveyed by color alone. The map uses MapLibre GL with an OpenStreetMap-based basemap, requires a WebGL-capable browser, and supports screens of 1280 × 800 pixels or larger.

5.5. User Control Interface

Table 7. User Control Interface

Page	User controls	Data shown
Risk Map	Pan/zoom; model run selector; toggle flagged network, HIN overlay, and crash points; search by street name	All sites colored by expected F&SI rate
Site Detail	Open from map or list; generate/regenerate memo; export memo (PDF)	Crash history by year, type, severity; EB estimate; top-5 SHAP factors; memo with citations
Backtest	Model run selector; method toggles (A, B, C)	Capture rate and 95% CI per method; capture-vs-network-share curve
Memos	Filter by status (draft, blocked, published); bulk export	Memo list with cost and grounding-check status
Ask (chat, stretch goal)	Free-text question box	Answer with citations to the records used

6. Communications / Device Interface Protocols

6.1. Network Communications

All traffic between browsers and the ATSC (IF-08) and between the ATSC and the TAMU AI Chat API (IF-07) uses HTTPS on TCP port 443 with TLS 1.2 or later; plain HTTP on port 80 redirects to HTTPS. The database port (5432) is reachable only on the internal Docker network. Timestamps are ISO 8601 in UTC; text is UTF-8.

6.2. Host Device (ATSC REST API, IF-08)

The backend exposes a versioned REST API under /api/v1. Requests and responses are JSON (RFC 8259) unless noted; map data is GeoJSON (RFC 7946). The API is self-documented as an OpenAPI 3.1 file served at /api/v1/openapi.json. Read endpoints are open to authenticated viewers; memo generation and chat require an editor role because they incur API charges.

Table 8. REST API Endpoints

Method	Path	Request	Response
GET	/api/v1/health	—	{"status", "db", "llm", "version"}
GET	/api/v1/runs	—	List of model runs: run_id, created, code_version, data_versions, published
GET	/api/v1/sites	Query: run_id, bbox (minLon,minLat,maxLon,maxLat), flagged	GeoJSON FeatureCollection (Table 9)
GET	/api/v1/sites/{site_id}	Query: run_id	Site detail: attributes, crash history, prediction, SHAP factors
GET	/api/v1/sites/{site_id}/memo	Query: run_id	Memo object (Section 6.4.3)
POST	/api/v1/sites/{site_id}/memo	{"run_id", "regenerate": bool}	202 Accepted with job_id
GET	/api/v1/jobs/{job_id}	—	{"status": queued | running | done | failed, "result"}
GET	/api/v1/backtest	Query: run_id	Per method: capture, ci_low, ci_high, n_test_fsi, flagged_miles
POST	/api/v1/chat (stretch goal)	{"question", "history"}	{"answer", "citations": [record references]}
GET	/api/v1/export/sites.{csv|geojson}	Query: run_id, flagged	File download (IF-11)
GET	/api/v1/sites/{site_id}/memo.pdf	Query: run_id	File download (IF-11)

Errors return an HTTP status code (400 invalid request, 401/403 not authorized, 404 not found, 409 memo blocked by grounding check, 429 rate limited, 503 LLM service unavailable) with a JSON body {"error": code, "message": text}.

6.3. Video Interface (Map Data Interface)

The /api/v1/sites endpoint returns one GeoJSON Feature per site. Geometry is a LineString (segment) or Point (intersection) in EPSG:4326. Feature properties are listed in Table 9.

Table 9. Site Feature Properties

Property	Type	Description
site_id	string	Stable site identifier
site_type	string	"segment" or "intersection"
road_name	string	Primary road name
length_mi	number	Segment length in miles (intersections: 250 ft influence zone)

Property	Type	Description
fsi_rate	number	Expected F&SI crashes per mile per year (ML-EB)
rank	integer	Rank within the run (1 = highest risk)
flagged	boolean	In the top 11% of network length
in_hin	boolean	Overlaps the consultant HIN
memo_status	string	none | draft | blocked | published

6.4. Device Peripheral Interface (LLM Service and Agent Tools)

6.4.1. IF-07 TAMU AI Chat API

The countermeasure agent (and the chat feature, if the stretch goal is delivered) calls the TAMU AI Chat API over HTTPS, authenticated with an API key issued through chat.tamu.ai and sent in the request header as documented at docs.tamus.ai. The key is read from an environment variable on the host and never sent to the browser. The model's name is a configuration setting, so the team can use a frontier model for memo writing and a lower-cost model for extraction.

Table 10. LLM Service Interface Parameters

Parameter	Value
Transport	HTTPS POST, JSON request and response
Authentication	API key from chat.tamu.ai (per docs.tamus.ai)
Tool use	Native function/tool calling if the endpoint supports it; otherwise the agent requests a JSON action object and the orchestrator executes the tool
Temperature	0.2 for memos; 0 for extraction
Timeout / retries	60 s per call; up to 3 retries with exponential backoff on 429 and 5xx
Logged per call	Timestamp, model, prompt hash, tool calls, input/output tokens, cost, latency, status
Data sent	Public data only: site attributes, crash aggregates, countermeasure records

6.4.2. IF-10 Agent Tool Interface

Tools are deterministic Python functions. The LLM chooses which tool to call and with which arguments; the tool computes the result; the orchestrator stores every result with the memo. The LLM never computes a number.

Table 11. Agent Tools

Tool	Arguments	Returns
get_crash_profile	site_id, years (default 2017–2022)	Totals by severity (K, A, B, C, O), by crash type, by light condition, and pedestrian/bike counts
get_site_context	site_id, run_id	Road attributes, EB estimate, rank, top-5 SHAP factors
search_countermeasures	site_id, crash_types, site_type	Matching library rows: cm_id, psc_name, cmf_id, cmf, star_rating, source_url
compute_benefit_cost	site_id, cm_id, run_id	Expected F&SI and total crashes reduced per year, present-value benefit, cost, B/C ratio, and the assumptions used

6.4.3. Memo Object

A memo is stored and returned as JSON with these fields: memo_id, site_id, run_id, status (draft, blocked, published), created, model, sections (crash profile, why flagged, recommended countermeasures, benefit/cost, references), tool_results (every tool call with

arguments and output), grounding_check (pass/fail and any unsupported numbers found), tokens_in, tokens_out, and cost_usd. A memo whose grounding check fails is stored with status "blocked" and is not shown as published.

6.5. Database Interface (IF-09)

The database is part of S1 (owner: Kabir Singh), which owns the schema. S2, S3, and S4 read and write the core tables in Table 12 through IF-09; no subsystem reads another subsystem's files directly.

Table 12. Core Database Tables

Table	Key fields
sites	site_id, site_type, geom, length_mi, road_name, attributes
crashes	crash_id, crash_date, severity, crash_type, geom, site_id, match_dist_ft, match_status
features	site_id, as_of_date, feature name/value pairs, source and source date
runs	run_id, created, code_version, data_versions, as_of_date, seed, hyperparameters, published
predictions	run_id, site_id, mu, k, eb_weight, n_eb, p_fsi, fsi_rate, rank, flagged
explanations	run_id, site_id, feature, shap_value, rank
backtest	run_id, method, capture, ci_low, ci_high, n_test_fsi, flagged_miles
memos	memo_id, site_id, run_id, status, content (JSON), tool_results, tokens, cost_usd
bit_log	test_run_id, timestamp, test, result, details

6.6. Exports and Operator Interface (IF-11, IF-12)

Exports are generated on demand: the ranked site list as CSV (RFC 4180, one row per site with the properties in Table 9) and GeoJSON, and memos as PDF with the run identifier and generation date in the footer. Operators run the pipeline from the worker container with the commands below; each command writes structured JSON logs and exits non-zero on failure.

Table 13. Operator Commands

Command	Function
atsc ingest --source <name> --path <file>	Validate and load a source (IF-01 to IF-06)
atsc network build	Build segments and intersections from IF-02/IF-03
atsc match	Map-match crashes to sites; report match rate
atsc features --as-of <date>	Compute features known as of the date
atsc train --run-id <id>	Train risk and severity models; compute EB and SHAP
atsc backtest --run-id <id>	Score methods A, B, C on the test window
atsc memos --run-id <id> [--flagged]	Generate memos in batch
atsc bit	Run the built-in test suite and write to bit_log
atsc publish / rollback --run-id <id>	Set or revert the run shown in the web application

AI TRAFFIC-SAFETY CONSULTANT

Juan Lopez
Jose Ruiz Garcia
Param Shelar
Kabir Singh

EXECUTION AND VALIDATION PLAN
AI CORPS – TEAM 4

REVISION – Draft
30 September 2026

EXECUTION AND VALIDATION PLAN
FOR
AI Traffic-Safety Consultant

PREPARED BY:

AI Corps #4 Oct 1, 2026

Author Date

APPROVED BY:

Param Shelar Oct 1, 2026

Project Leader Date

John Lusher II, P.E. Date

T/A Date

Change Record

Table 1 List of changes

Rev.	Date	Originator	Approvals	Description
-	9/30/2026	Juan Lopez, Jose Ruiz Garcia, Param Shelar, Kabir Singh		Draft Release
-	10/1/2026	Param Shelar		Named subsystem owners throughout; owner column added to the verification matrix; aligned with ConOps Rev C and the FSR hosting footprint
-	10/1/2026	Param Shelar		Scope reduction: chat page moved to a stretch goal, NVIDIA Jetson dropped, memo export PDF only, fixed intersection rule (three or more legs, 250 ft zone)

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1. Overview

This Execution and Validation Plan describes how AI Corps Team 4 will build the AI Traffic-Safety Consultant (ATSC) between October 2026 and May 2027, and how the team will prove that the finished system meets the Functional System Requirements (FSR). It has two parts:

- **Execution plan (Section 3):** who does what and when: team roles, schedule and milestones, work breakdown, the go/no-go decision, resources, risks, and communication.
- **Validation plan (Section 4):** how each requirement is verified: the backtest protocol for the risk model, grading of the countermeasure memos, software and fault-injection tests, user acceptance testing with city staff, and a requirements verification matrix that traces every FSR requirement to a test.

The plan is written so that a result the team did not hope for, such as the model failing to beat the consultant's High-Injury Network, is still a completed and reportable validation outcome.

2. References and Definitions

2.1. References

ATSC Functional System Requirements (FSR)
Requirements verified by this plan
Revision – (Draft), 30 Sep 2026
ATSC Interface Control Document (ICD)
Interface definitions referenced by the tests
Revision – (Draft), 30 Sep 2026
Team 4 Concept of Operations (CONOPS)
Operational concept
Revision C, 1 Oct 2026
Bryan/College Station MPO Comprehensive Safety Action Plan
Source of the consultant High-Injury Network (comparison baseline)
Study began Oct 2023
AASHTO Highway Safety Manual
Empirical Bayes method, KABCO scale
1st Edition, 2010
Zheng et al., Judging LLM-as-a-Judge (arXiv:2306.05685)
Practice for LLM-assisted grading and its biases
2023

2.2. Definitions

Each requirement is verified by one of four standard methods:

Inspection (I)	Examination of code, documents, data, or outputs against the requirement without running a test.
Analysis (A)	Evaluation by calculation or statistics on recorded results (for example, the backtest).
Demonstration (D)	Operation of the system to show a function works, judged pass/fail by observation.
Test (T)	Controlled, repeatable measurement against a numeric pass criterion.
Capture rate	Share of 2023–2025 F&SI crashes that occurred on sites flagged by a method.
Go/no-go	A scheduled decision point where the team continues, changes, or narrows scope based on evidence.
Seeded fault	An error deliberately inserted to check that the built-in tests catch it.

3. Execution Plan

3.1. Objective and Success Criteria

The project answers one question for one class of consultant deliverable: can an LLM-based consulting tool reproduce the traffic-safety prioritization and countermeasure memos a consultant delivers, at comparable quality and substantially lower cost? The project succeeds when all of the following are true:

1. The backtest of methods A (consultant HIN), B (negative binomial with EB), and C (ATSC ML-EB) is complete, with capture rates and confidence intervals reported whatever they show.
2. Countermeasure memos are generated for the flagged sites, pass the automated grounding check, and are blind-graded by a traffic-safety reviewer.
3. The web application is deployed and passes user acceptance testing with city staff.
4. Cost per memo and total API cost are documented next to the consultant fee benchmark.
5. Results are presented at City Hall (fall) and to City Council (spring).

3.2. Team Organization

Table 2. Team Roles and Responsibilities

Role	Person	Responsibilities
Team leader	Param Shelar	Schedule, requirement changes, client contact, integration, verification matrix (Section 4.6), final reports
S1 Road Network, Features, and Database	Kabir Singh	IF-01 to IF-05 ingest, network build, map-matching, feature store, database (IF-09), data validation; ICD custodian
S2 Risk Model and Backtest	Param Shelar	Methods A, B, C; EB; severity model; SHAP; backtest and bootstrap; model card
S3 Countermeasure Agent	Jose Ruiz Garcia	IF-06 library, agent tools (IF-10), LLM interface (IF-07), prompts, grounding check, cost logging
S4 Web Application and Deployment	Juan Lopez	FastAPI backend (IF-08), React/MapLibre frontend, exports (IF-11), containers and hosting; City liaison for reviewer recruitment and acceptance sessions
Client	City of College Station (Sam Rivera)	Requirements approval, data contacts, city staff for acceptance testing
Course T/A	Roman Venegas (course); Sabyasachi Gupta (AI program T/A)	Technical review; involved at least every other meeting
Memo reviewer	TTI safety researcher (to be recruited)	Blind grading of memos (Section 4.3)

Every team member also owns the unit tests for their own subsystem. The team leader owns the verification matrix (Section 4.6) and signs off each milestone.

3.3. Schedule and Milestones

Figure 1 shows the planned schedule, colored by owning subsystem. Dates after M1 are targets; City Hall and City Council dates will be set with the City. The fall semester delivers the data pipeline, risk model, and backtest; the spring semester delivers the agent, web application, and validation.

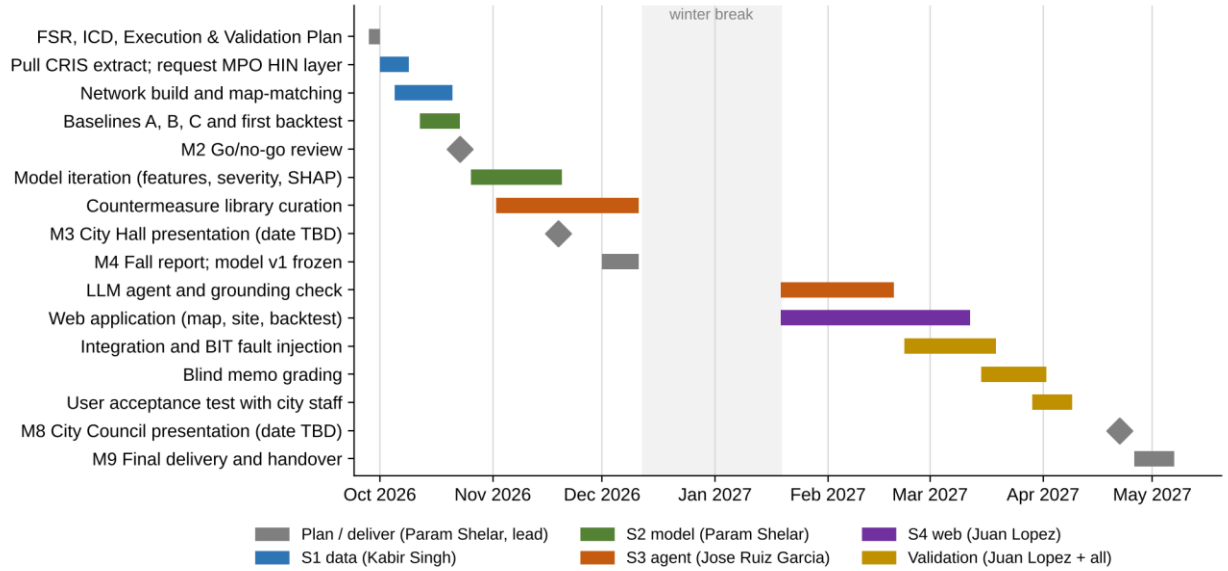


Figure 1. Project Schedule

Table 3. Milestones and Exit Criteria

ID	Milestone	Target date	Exit criteria
M0	FSR, ICD, and this plan submitted	1 Oct 2026	Documents submitted; requirements baselined at Revision –
M1	Data acquired	9 Oct 2026	CRIS 2015–2025 extract ingested and validated; HIN layer received or request escalated
M2	Go/no-go	23 Oct 2026	Methods A, B, C run end-to-end; first capture-rate results (Section 3.5)
M3	City Hall presentation	Nov 2026 (TBD)	Backtest results and risk map shown to city staff; feedback recorded
M4	Model v1 frozen	11 Dec 2026	Final backtest with bootstrap CIs; model card drafted; fall report submitted
M5	Agent MVP	19 Feb 2027	Memos generated for all flagged sites; grounding check blocks seeded invented numbers
M6	Web beta	12 Mar 2027	Risk map, site detail, and backtest pages working on the host; exports working
M7	Validation complete	9 Apr 2027	Blind grading, BIT fault injection, performance, and user acceptance tests reported
M8	City Council presentation	Apr 2027 (TBD)	Quality and cost comparison presented
M9	Final delivery	7 May 2027	Support items in FSR Section 4 delivered; handover meeting held

3.4. Work Breakdown

Table 4. Work Breakdown

Workstream	Main tasks	Output
1. Road network and data (S1, Kabir Singh)	Download CRIS extracts; load roadway inventory and OSM; build 0.1-mile segments and intersections; map-match crashes; feature tables with as-of dates; validation checks	Validated site and crash tables; match-rate report
2. Models and backtest (S2, Param Shelar)	HIN import (A); NB SPF + EB (B); Poisson GBM + overdispersion + EB + severity (C); SHAP; flag top 11%; capture rate with bootstrap; sensitivity at other network shares	Ranked sites per method; backtest report; model card

Workstream	Main tasks	Output
3. LLM agent (S3, Jose Ruiz Garcia)	Curate countermeasure library; implement four tools; prompts and memo template; grounding and citation checks; cost and latency logging	Memo generator; library CSV; memo batch for flagged sites
4. Web application (S4, Juan Lopez)	REST API per ICD; risk map, site detail, and backtest pages; CSV, GeoJSON, and PDF exports; containers and deployment; chat page only as a stretch goal after M6	Deployed web application; deployment guide
5. Validation and delivery (Param Shelar lead; Juan Lopez sessions)	Test suites; fault injection; reviewer recruitment and grading; user acceptance test; documentation; presentations	Verification report; user guide; runbook; presentations

3.5. Go/No-Go Decision (M2)

The first three weeks exist to find out early whether the approach works. At M2 the team reviews the first backtest and takes one of three paths:

Table 5. Go/No-Go Outcomes

Result at M2	Decision
Method C or B captures more test-window F&SI crashes than the HIN (A)	Go: continue to model iteration as planned.
Neither B nor C beats A, but pipeline and backtest are sound	Go with changed framing: the finding that EB does not improve on the HIN is reported; effort shifts toward the countermeasure agent, and the HIN itself becomes the site list for memos.
HIN layer or CRIS data not obtained	Hold: team leader escalates through Sam Rivera and the MPO; meanwhile compare B and C with each other and reconstruct the HIN from its published description, noting the limitation.

3.6. Resources and Budget

- **Data:** all inputs are public (TxDOT CRIS, TxDOT roadway inventory, OpenStreetMap, MPO and city GIS); no purchase required.
- **LLM API:** TAMU AI Chat API through tokens; charges are covered by the program, and the team must get approval before exceeding the approved budget. Every call is logged with its cost, and the team leader reviews the running total at each biweekly meeting.
- **Compute:** one Linux host meeting the FSR hosting footprint (4 CPU cores, 8 GB RAM, 128 GB storage, no GPU); a Raspberry Pi 5 is the reference test host.
- **Software:** open-source only (Python, PostgreSQL/PostGIS, FastAPI, React, MapLibre).
- **People:** one traffic-safety reviewer for about four hours of blind grading, and two to four city staff for a one-hour acceptance session.

3.7. Risk Management

Table 6. Risk Register

Risk	Likelihood	Impact	Mitigation	Trigger / owner
Model does not beat the HIN	Medium	Medium	Run backtest first (M2); report EB-vs-HIN as a valid result	M2 result / Param Shelar
MPO HIN GIS layer not provided	Medium	High	Request in week 1; escalate via City; reconstruct from published maps	No reply by 9 Oct / Param Shelar
Crash location noise	High	Medium	Generous, logged match tolerance; severity weighting; report match rate	Match rate < 95% / Kabir Singh

Risk	Likelihood	Impact	Mitigation	Trigger / owner
Data leakage invalidates backtest	Low	High	As-of dates on every feature; automated cutoff check; peer review of feature list	BIT failure / Param Shelar
LLM invents numbers or credentials	Medium	High	Tools compute all numbers; grounding check blocks memo; human review	Blocked memos / Jose Ruiz Garcia
Host too small for the full stack	Low	Medium	Container resource caps (ICD Table 2); CPU-only model; full-stack test on the Raspberry Pi 5 before M6	Out-of-memory or > 4 h run / Juan Lopez
API budget overrun	Low	Medium	Per-memo cost log; cheaper model for extraction; approval before exceeding	75% of budget / Param Shelar
Reviewer not available	Medium	Medium	Recruit through TTI by January; fallback: faculty reviewer plus calibrated LLM judge	No reviewer by 1 Feb / Juan Lopez
SS4A federal program ends (IIJA expires 30 Sep 2026)	High	Low	Frame the tool as prioritization for the City's own capital budget, not grant writing	N/A / Param Shelar
Team is about 10 days behind; competing with Team 1	High	Medium	Narrow scope to one deliverable; weekly task-list review	Missed milestone / Param Shelar

3.8. Communication and Reporting

- Team meeting at least every other week, with the T/A involved; decisions and requirement changes recorded in the repository.
- Task tracking and code in the team GitHub repository; every change runs the automated test suite.
- Client updates to Sam Rivera at each milestone (M1 to M9), and before any change to the FSR.
- A short status note at each milestone: done, next, risks, and API spend to date.

4. Validation Plan

4.1. Verification Approach

Verification runs at three levels: (1) automated unit and integration tests on every code change; (2) milestone verification events (backtest, fault injection, performance, grading, and acceptance) run against a frozen model run and software version; and (3) client acceptance. Each FSR requirement is assigned a verification method and pass criterion in the matrix in Section 4.6. A requirement is verified only when its evidence (log, report, or signed test record) is stored in the repository.

4.2. Risk Model Validation (Backtest Protocol)

The backtest is the main evidence for the project. It follows the split in Figure 2: all three methods rank sites using only information available by 31 December 2022, and are scored on crashes that happened in 2023–2025.



Figure 2. Backtest Data Split

1. **Freeze inputs.** Record the version and as-of date of every data source; run the leakage check (FSR 3.2.1.5). Any feature dated after 31 Dec 2022 stops the run.
2. **Rank.** Produce a ranking for each method: A = consultant HIN as published; B = negative binomial SPF with EB; C = ATSC ML-EB. Flag the top 11% of network length for B and C; use the HIN as published for A and record its actual share of network length.
3. **Score.** Count 2023–2025 F&SI crashes (KABCO K and A) on flagged sites and divide by all 2023–2025 F&SI crashes in the analysis area to get the capture rate C for each method.
4. **Uncertainty.** Resample the test-window F&SI crashes with replacement 1,000 times, recompute each method's capture rate and the paired differences C−A and B−A, and report the 2.5th and 97.5th percentiles as 95% confidence intervals.
5. **Fairness check.** If the HIN covers a share of network length different from 11%, also compare B and C at the HIN's exact share, so no method gains from flagging more road.
6. **Sensitivity.** Repeat scoring at 5% and 20% of network length and with total crashes instead of F&SI.
7. **Report.** Publish all results on the backtest page and in the model card, including results that favor the HIN.

With roughly 100 F&SI crashes per year in Brazos County, the 2023–2025 window provides about 300 test crashes, enough for the bootstrap intervals to distinguish differences of several percentage points in capture rate.

4.3. Countermeasure Memo Validation

4.3.1. Automated Checks

Every memo passes three automated checks before it can be published: (1) numeric grounding: every number in the text must match a value in the memo's stored tool results; (2) citation check: every recommended countermeasure must carry a CMF Clearinghouse ID or FHWA source present in the library; (3) completeness: all required sections are present. A seeded set of 20 memos containing invented numbers or citations is used to confirm the checks work.

4.3.2. Blind Expert Grading

- **Sample:** 15 flagged sites chosen at random, stratified by site type (segments, signalized and unsignalized intersections).
- **Comparison memos:** for each site, a reference memo written in consultant style from the CSAP project recommendations where the CSAP covers that site, otherwise by a team member following the same template without AI assistance.
- **Blinding:** both memos use the same template and formatting, are labeled X and Y at random, and are graded without knowing which is AI-generated.
- **Rubric:** six criteria scored 1–5 (Appendix A). Pass: ATSC mean ≥ 3.5 and within 0.5 point of the reference mean (FSR 3.2.1.16).
- **Optional LLM judge:** a second, larger set may be scored by an LLM judge with randomized order and length normalization, used only if its scores agree with the human reviewer on the 15-site set.

4.3.3. Cost Comparison

For each memo the logged token cost and generation time are summed and reported next to the consultant benchmark (about $200,000 for a comparable Kimley-Horn safety action plan). The report states plainly that the tool reproduces the prioritization and memo portions of such a plan, not its public engagement or engineering design work.

4.4. Software and System Tests

Table 7. Software and System Tests

Test	Method	Pass criterion
Unit tests	Automated on every change (pytest, frontend test runner)	All pass; coverage ≥ 70% on S1–S3 code
Interface tests	Contract tests for each ICD endpoint and file format	Responses match the OpenAPI schema and ICD tables
BIT fault injection	20 seeded critical faults: shifted dates, swapped lat/lon, post-cutoff feature, duplicated crashes, invented memo number, missing citation, etc.	≥ 19 of 20 detected (FSR 3.2.5.1.1.1)
BIT false alarms	Run BIT on 20 known-good runs	≤ 1 false alarm (< 5%, FSR 3.2.5.1.1.2)
Reproducibility	Re-run a stored run ID from scratch	Identical ranking (FSR 3.2.1.21)
Performance	Scripted browser test, 5 concurrent users, campus network, 10 trials	Map ≤ 5 s and site page ≤ 2 s in 9 of 10 trials
Resilience	Block LLM API during a session; fail a model run mid-way	Map, sites, backtest stay up; memos "pending"; last run kept (FSR 3.2.5.1.2)
Accessibility	Automated scan (axe or Lighthouse) plus keyboard-only walkthrough	No critical WCAG 2.1 AA violations

Test	Method	Pass criterion
Security	Inspect repository and browser traffic for keys; check TLS	No key in code or browser; TLS 1.2+ only

4.5. User Acceptance Testing

Two to four City of College Station staff will use the deployed system for about one hour, with a team member observing. Each participant completes five tasks: find the highest-risk site near a given street; explain why a site was flagged; open and export a memo; compare the model with the HIN on the backtest page; and export the ranked site list for the City GIS. Acceptance requires that at least 80% of tasks are completed without help and that the client representative confirms in writing that the system meets the intent of the FSR. Issues found are logged and fixed or documented before M9.

4.6. Requirements Verification Matrix

Table 8 traces every FSR requirement to its verification method, pass criterion, owner, and milestone. The owner produces the evidence for the requirement: S1 Kabir Singh, S2 Param Shelar, S3 Jose Ruiz Garcia, S4 Juan Lopez; "Lead" items are integrated and run by the team leader with input from every owner. Owners match the allocation in FSR Table 4. The method codes are defined in Section 2.2.

Table 8. Requirements Verification Matrix

FSR §	Requirement	Method	Pass criterion	Owner	MS
3.2.1.1	Network Coverage	I, T	Every site in the network table has a prediction; count check = 0 missing	S2	M2
3.2.1.2	Site Definition	T	Segment length distribution within 0.05–0.15 mi; every node with ≥ 3 legs is an intersection site with a 250 ft zone	S1	M1
3.2.1.3	Crash Data Ingest	I	All years 2015–2025 loaded; field checks pass	S1	M1
3.2.1.4	Map-Matching Rate	T	≥ 95% of geocoded F&SI crashes matched; unmatched log complete	S1	M1
3.2.1.5	Temporal Leakage Control	T	Seeded post-cutoff feature blocks the run	S1	M2
3.2.1.6	Comparison Methods	D	A, B, C produce rankings on identical inputs	S2	M2
3.2.1.7	Empirical Bayes Correction	I, A	Code review; weights in (0,1); hand-check of toy example	S2	M2
3.2.1.8	Severity Estimate	A	Severity model calibration curve reported	S2	M4
3.2.1.9	High-Risk Flagging	T	Flagged length = 11% ± 0.5% of network	S2	M2
3.2.1.10	Backtest Reporting	A, D	Capture rates with 1,000-resample CIs shown for A, B, C	S2	M4
3.2.1.11	Predictive Performance Goal (should)	A	C vs. A difference and CI reported (goal, not pass/fail)	S2	M4
3.2.1.12	Prediction Explanations	D	Top-5 factors shown for every flagged site	S2	M4
3.2.1.13	Countermeasure Memo Content	I	Completeness check passes on all memos	S3	M5
3.2.1.14	Numeric Grounding	T	All 20 seeded invented-number memos blocked	S3	M5
3.2.1.15	Citation Grounding	T	All seeded missing-citation memos blocked	S3	M5
3.2.1.16	Memo Quality Goal (should)	A	Mean ≥ 3.5/5 and within 0.5 of reference	S3	M7
3.2.1.17	Cost and Time per Memo	T	Cost logged for 100% of memos; median ≤ 2 min and ≤ $1.00	S3	M5
3.2.1.18	Web Application Functions	D	Map, site detail, and backtest pages demonstrated in acceptance test; chat page only if delivered	S4	M6
3.2.1.19	Web Responsiveness	T	Performance test in Table 7	S4	M7
3.2.1.20	Exports	D	CSV, GeoJSON, and PDF files open correctly	S4	M6
3.2.1.21	Reproducibility	T	Re-run gives identical ranking	S2	M4

FSR §	Requirement	Method	Pass criterion	Owner	MS
3.2.2.1	Hosting Footprint	I	Container limits reviewed; full stack runs on the Raspberry Pi 5 reference host	S4	M6
3.2.2.2	Model Training Time	T	Full run ≤ 4 h on the reference host, timed from logs	S2	M4
3.2.2.3	Deployment	D	Clean deploy from images and config by a non-author	S4	M9
3.2.3.1.1	Input Data Sources	I	Each ICD source ingested in its defined format	S1	M1
3.2.3.1.2	Input Validation	T	File with 3% bad records stops the pipeline	S1	M1
3.2.3.1.3	External Commands	I	ICD lists every endpoint and command in the code	Lead	M6
3.2.3.2.1	Data Output	T	Interface contract tests pass	S4	M6
3.2.3.2.2	Diagnostic Output	I	Logs present for each stage; 90-day retention configured	Lead	M6
3.2.3.2.3	Model Artifact Output	D	Artifacts exported and reloaded	S2	M4
3.2.3.3	Network Interfaces	T	TLS scan: 1.2+ only; HTTP redirects	S4	M6
3.2.4.1	Server Environment	I	Versions confirmed on host	S4	M6
3.2.4.2	Client Environment	D	Walkthrough in four browsers, desktop and tablet	S4	M7
3.2.4.3	Accessibility (should)	T	No critical WCAG 2.1 AA violations	S4	M7
3.2.4.4	Data Classification	I	Data inventory confirms public sources only	S1	M6
3.2.4.5	Credential Handling	I, T	Secret scan of repository; browser traffic check	S4	M6
3.2.5.1.1.1	BIT Critical Fault Detection	T	≥ 19 of 20 seeded faults detected	Lead	M7
3.2.5.1.1.2	BIT False Alarms	T	≤ 1 false alarm in 20 good runs	Lead	M7
3.2.5.1.1.3	BIT Log	D	Results retrievable from operator interface	Lead	M7
3.2.5.1.2	Isolation and Recovery	T	Resilience test in Table 7	S4	M7

4.7. Test Reporting and Anomaly Handling

Each verification event produces a short test record: requirement(s), software version and model run ID, procedure, raw results, pass/fail, and tester. Failures are logged as issues in the repository with severity (critical: invalidates results or exposes data; major: requirement not met; minor: cosmetic). Critical and major issues must be closed and the affected tests re-run before the next milestone. If a requirement cannot be met, the team leader proposes a change through the FSR change process with the client's approval rather than leaving it unverified.

Appendix A: Memo Grading Rubric

Table 9. Memo Grading Rubric

Criterion	What the reviewer checks	1 (poor)	5 (excellent)
Analysis	Crash patterns correctly read and linked to the countermeasures	Generic or wrong reading	Specific, correct diagnosis of the crash pattern
Coverage	Relevant countermeasure options considered	One obvious option only	Appropriate range, including low-cost options
Grounding	Numbers and claims traceable to data and sources	Unsupported claims	Every claim sourced and correct
Actionability	A city engineer could scope a project from it	Vague	Clear next steps, cost, and B/C
Presentation	Clear, well organized, appropriate length	Hard to follow	Ready to share with Council
Scope compliance	Stays within a planning memo; flags need for engineering study	Overreaches or omits limits	Scope and limits stated plainly